

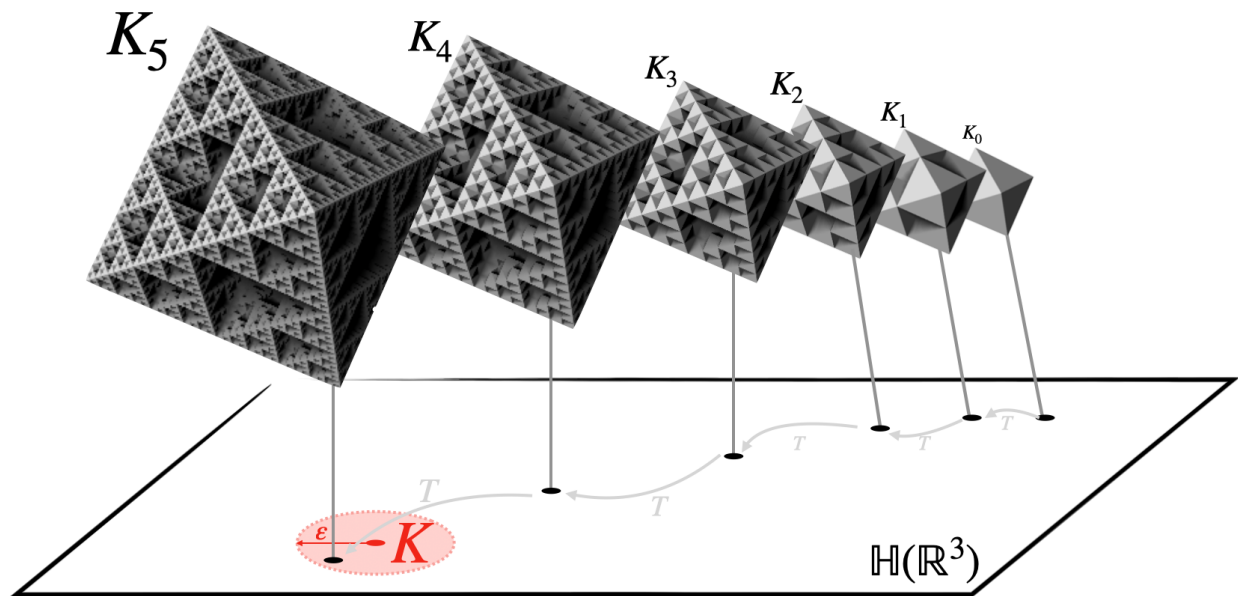
Attractors of Iterated Function Systems: where do they live and how to see them

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Aquesta dissertació està dedicada a la meua família, entre ells: el meu avi Carlos i el meu abuelito José, els quals van vetllar incansablement per mí cada dia de la meua infància; la meua yaya Emma, qui em va ensenyar a sumar i restar; la meua abuelita Rosario, qui em va ensenyar a llegir; la meua tieta Sila, que va ser per mí com una segona mare i el meu tiet Aitala, am qui he viscut aventures que mai oblidaré. No obstant això, aquesta dissertació està especialment dedicada als meus pares, Iván i Lara, perquè gràcies als seus sacrificis, estic complint el meu somni.

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Abstract

In this project, we will develop the elementary theory of Iterated Function Systems (IFS) and show one of its main results, the existence and uniqueness of attractors: sets that are invariant under an IFS. We will then proceed to explore methods to estimate the distance between a point and an attractor, with the goal of visualising them via the Ray-Marching algorithm. We will also introduce our method of Ray-Marching attractors.

Unless otherwise stated, all diagrams of this project have been produced by the author. Most diagrams were produced using the $\text{\LaTeX}$ package `tikz-cd` or coded in High Level Shading Language with the methods described throughout this project.

1 Introduction

1.1 The goal of The Project

Fractals are objects that elude definitions, citing [Bar93, Page 34]: *it is better not to think of fractals as defined by a short legalistic statement, but by the many pictures and contexts that refer to them.* It is hard to capture all the meaning we think a fractal should have: intricate structure, self-similarity, “infinite detail”, etc. One way we can mathematically approach the study of these sets is to think of fractals as sets in a metric space that are invariant under a certain kind of maps, called Iterated Function Systems (IFS). This notion captures classic examples of sets that we may think of as being fractal sets and provides a good structure to work with them. The goal of this project will be to visualise these sets with the aid of computers.

Example 1.1 The fractal commonly referred to as Menger’s Sponge, which we denote by $\mathcal{M}$, is an example of a set that arises as a fixed point of a contractive IFS. However, many other strange fractals can also be obtained by means of an IFS. See Figure 1 for an example.

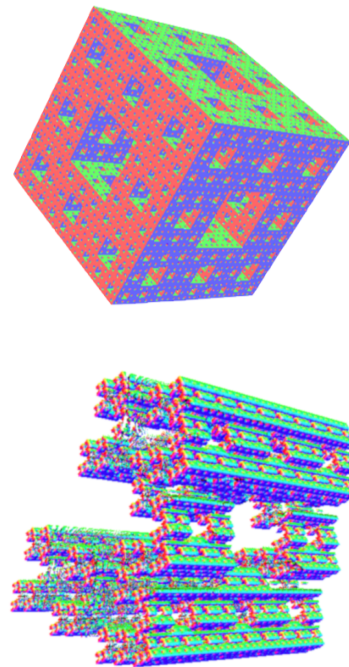


Figure 1: Menger’s Sponge, $\mathcal{M}$, and a “broken version” of it

Ever since the introduction of IFS Theory for the construction of fractal sets by Hutchinson [Hut81], it has become natural for the mathematician to ask how one can visualise these sets to observe their beautiful structure. Initial methods for this were rather rudimentary, such as the so-called Chaos Game, or the Deterministic Algorithm, described in [Bar93, Chapter 3]. These methods revolve around approximating the fractal set itself, i.e. finding points inside the fractal, and then plotting the points as pixels on a screen. These algorithms quickly became obsolete, since their memory requirements grow rapidly as one aims for higher resolutions. A wide range of other solutions, which we will come back to later, have been proposed, such as making use of the very rich concept of invariant measures, which are measures supported on the fractal set (see [Bar93, Chapter 9]): pixels on the screen are then interpreted as some open set in space and they are coloured according to their measure. Another method is the Escape Time Algorithm, traditionally used for visualising Julia Sets, but also studied in the context of IFS [Bar93, Chapter 7], [HPS90, Section 4]. The ideas to visualise fractal sets that we discuss in this project are of a different nature: unlike the methods described above, which were initially thought for 2-D visualisation, we will be focusing on the 3-D visualisation of fractals, introduced in [HSK89], which became an area of interest once it was observed that the fractals that arise from the dynamics of quadratic functions on the complex plane have four-dimensional analogs in the space of quaternions [Nor98, Chapter 33]. Since then, 3-D fractal visualisation has become a very popular topic amongst both mathematicians and computer graphics enthusiasts, with a variety of solutions having been used to produce images of fractals. The algorithm we will be studying in depth in this project is Ray-Marching, recently popularised in the computer graphics circles by Inigo Quilez [Quib],

and first used to Ray-March 3-D IFS Fractals by Hepting, Prusinkiewicz, and Saupe [HPS90]. We will be studying this algorithm because, despite its origins in computer graphics, it is of mathematical interest since it requires us to understand in depth the question of estimating distances between a given point in a metric space and a given set. When the set in question is a fractal object, this suddenly becomes a very rich topic of study, which inspired many different solutions, ranging from basic deterministic methods to methods based on dynamical systems, and even measure and ergodic theory!

1.2 Introduction to Ray-Marching

Ancient Greek philosophers proposed the theory of extramission, which held that visual perception is obtained through rays of light emitted by our eyes, which travel in straight lines flowing out of the pupil. These rays were thought to touch objects and generate a medium between the object and the viewer which allowed aspects of the object to contact the soul. [Lin71]

This theory offers inspiration for how we might construct an algorithm to visualize a set Σ in space using a computer. One of the main algorithms used for this is Ray-Marching, which as the name suggests, marches (moves in a discrete manner) a “ray of light” in some specified direction and detects, up to some specified tolerance, if this ray of light ends up colliding with Σ or if the light ray escapes to infinity. The pixels on the screen are then coloured accordingly.



Figure 2: How Plato would have seen a Psykter according to Euclid. Photo credits [Ngu], [Dal].

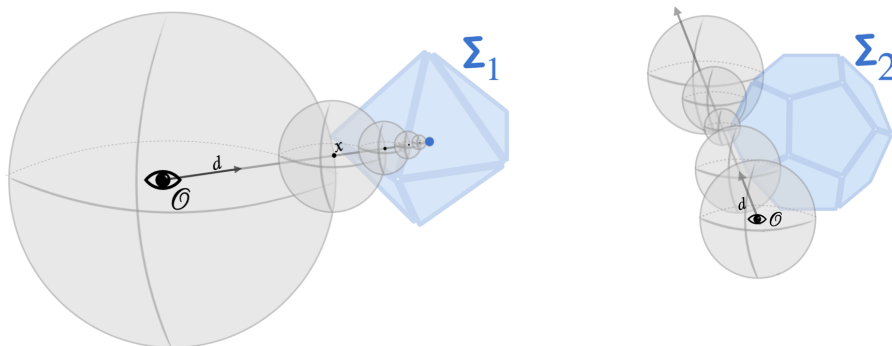


Figure 3: On the left, a ray is marched in direction d and collides with Σ_1 . On the right, a ray being marched in direction d escapes to infinity.

Algorithm 1 Ray-March a set Σ in $\mathbb{R}^3$

Input: $0 < \epsilon < E \in \mathbb{R}$ (collision and escape tolerances); $\mathcal{O} \in \mathbb{R}^3$ (ray-origin); $\Sigma \neq \emptyset$ a set.

```
for every pixel  $p_{ij}$  on the screen do
  Calculate ray direction  $d \in \mathbb{R}^3$ 
  Let  $x = \mathcal{O}$ 
  Let  $\rho$  be the Euclidean distance between  $x$  and  $\Sigma$ 

  while  $\epsilon < \rho < E$  do
    Move  $x \rightarrow x' = x + \rho d$ 
    Calculate  $\rho$ 
  end while

  if  $\rho < \epsilon$  then
    Colour pixel  $p_{ij}$  with “set colour”
  else
    Colour pixel  $p_{ij}$  with “background colour”
  end if
end for
```

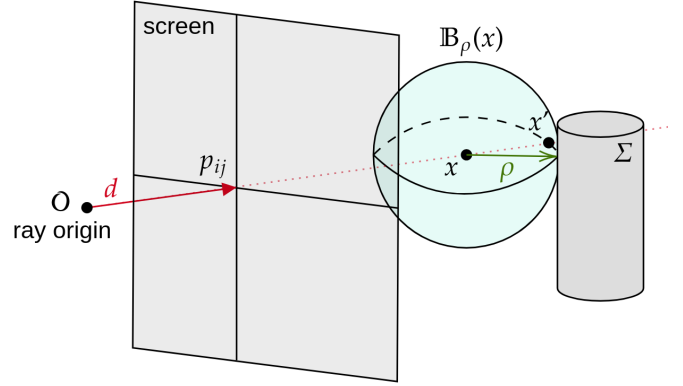


Figure 4: Diagram illustrating the Ray-Marching algorithm.

The screen is thought of as a plane in space and ray directions are calculated accordingly, then a ray is marched in the specified direction until it hits the object or escapes. The problem with this algorithm is that it relies on some method for you to compute the Euclidean Distance between a point and a set to detect these collisions or escapes. These distance functions can be computed for some “nice” sets, like cubes, tetrahedrons, octahedrons, spheres, etc. But as soon as the set becomes more complicated, such as for a fractal set, this task becomes impossible, and one has to resort to approximating the distance function. We refer to this as the **distance estimation problem**. The fundamental idea of this project will be to study some methods that can be used to overcome the distance estimation problem with the end goal of Ray-Marching fractal sets.

The outline of the project is as follows: in Chapters 2 and 3, we introduce the basic preliminaries needed: a quick brush-up on the theory of metric spaces and an introduction to the crucial notion of the Hausdorff metric and the space of non-empty compact subsets $\mathbb{H}(S)$. Some basic results will be presented regarding these concepts and then we shall introduce the elementary ideas of IFS Theory, which culminates with “The Attractor Theorem”, which guarantees the existence of fractals in the sense that we have defined them. We will then move forward to studying the distance estimation problem for fractal sets arising from an IFS, giving an overview of the methods in the literature and showcasing how the different methods produce different results. Finally, we will introduce our method of Ray-Marching IFS fractals, which exploits the Ray-Marching algorithm itself to overcome the distance estimation problem. As we will see, this algorithm resembles one of the methods discussed in the literature review, but provides slightly faster convergence to the fractal and as a result, better graphical results.

2 Preliminary background

2.1 Metric Spaces

We start the exploration by recalling some basic notions of the theory of metric spaces, which is the theoretical underpinning of most of the content of this project.

Definition 2.1 A metric space is a pair (S, ρ) where S is a non-empty set and $\rho : S \times S \rightarrow \mathbb{R}$ is a function, called the metric, that satisfies:

1. ρ is non-negative and $\rho(x, y) = 0$ if and only if $x = y$.
2. ρ is symmetric, i.e: for all $x, y \in S$, $\rho(x, y) = \rho(y, x)$.
3. ρ satisfies the triangle inequality, i.e: for all $x, y, z \in S$, $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$.

The elements of S are referred to as points.

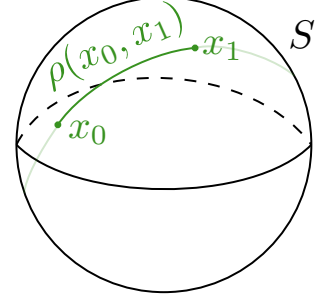


Figure 5: Metric spaces have geometric meaning, but they can appear where you less expect them to be!

From now on, we use (S, ρ) to denote a metric space and whenever we write S or ρ we refer to a space or its metric.

Example 2.2 The canonical example of a metric space, and the one whose sets are of interests to us is the Euclidean space $(\mathbb{R}^n, d_e)$, where $d_e(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{\frac{1}{2}}$.

Definition 2.3 We use the notation $\mathbb{B}_r(x)$ to denote the open r -neighborhood of x : $\{y \in S \mid \rho(x, y) < r\}$. Similarly, we denote $\mathbb{B}_r[x]$ to be the closed r -neighborhood of x : $\{y \in S \mid \rho(x, y) \leq r\}$.

The fundamental object of study for us will be the sets that live in metric spaces, and the transformations that act on those sets. As explained by the introductory discussion on Ray-Marching, the majority of the discussion in this project will revolve around the idea of finding distances between points and sets:

Definition 2.4 Let $x \in S$ be a point and $\Sigma \subseteq S$ be a non-empty set. The **distance function** of Σ is the function $S \rightarrow \mathbb{R}$ defined by:

$$\rho(x; \Sigma) := \inf_{y \in \Sigma} \rho(x, y)$$

Remark 2.5 Many of the distance functions we will use in this project are calculated algorithmically. We will repeatedly refer to Inigo Quilez's list of distance functions: [Quia].

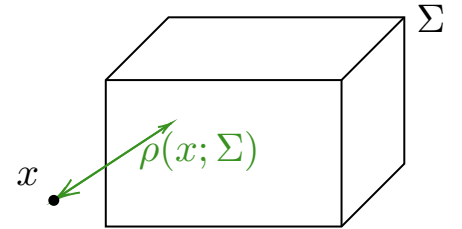
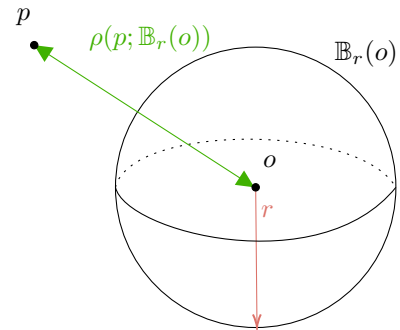


Figure 6: Illustration of the definition of the distance function of a set.

Example 2.6 Distance functions are easy to compute for some “nice” sets. For example, the distance function to the a ball (open or closed) of radius r centered at a point $o \in S$ is given by:

$$\rho(x; \mathbb{B}_r(o)) = \max\{0, \rho(x, o) - r\}$$



Another concept which we will need use the discussions to follow is that of the diameter of a set:

Definition 2.7 Let $\Sigma \subseteq S$ be non-empty. The **diameter** of Σ , is defined to be:

$$\text{diam}(\Sigma) := \sup_{x, y \in \Sigma} \rho(x, y)$$

2.2 The space where fractals live

The important mechanisms we will study don’t take place in metric spaces whose points are the traditional idea of a point, for example, coordinates in $\mathbb{R}^3$. Rather, the mechanisms will take place in a space where points are subsets of other spaces. Fractals, *live* as points of the following space:

Definition 2.8 Let (S, ρ) be a metric space. $\mathbb{H}(S)$ is the set of non-empty, compact subsets of S .

A natural question to ask oneself is whether we can come up with a function that describes the distance of points in this space $\mathbb{H}(S)$. The idea of the distance from a point to a set can give inspiration to construct this function:

Definition 2.9 Let $A, B \in \mathbb{H}(S)$. The Hausdorff distance between A, B is

$$d_{\mathbb{H}}(A, B) := \max \left\{ \sup_{x \in A} \rho(x; B), \sup_{x \in B} \rho(x; A) \right\}$$

Which is just the maximum distance you “can be forced to travel” from either of the two sets until you reach the other. There is a very natural way to think about the Hausdorff distance, citing Barnsley [Bar93, Page 289]: *in some sense, the Hausdorff distance is the distance of perception: what we call a small change in the appearance of a picture is probably a small change in Hausdorff distance, a continuous growth in terms of the context of botany, or a continuous change in the context of a chemical system can be often replaced, pedantically, by “continuous in the Hausdorff metric”.*

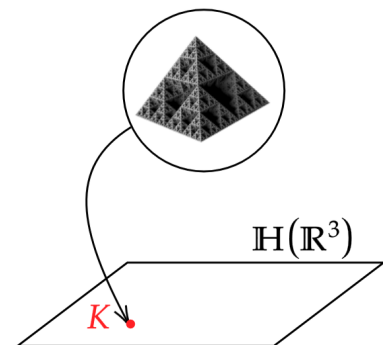


Figure 7: Fractals are “interesting points” in the space $\mathbb{H}(S)$.

So far I have been using the words “Hausdorff distance” and “Hausdorff metric” without giving any justification whether it actually constitutes a metric space along with $\mathbb{H}(S)$. Before proving this, let us look at a heuristic example to get a better feel for what this distance does.

Example 2.10 Let $(S, \rho) = (\mathbb{R}^2, d_e)$ and consider the points of $\mathbb{H}(\mathbb{R}^2)$ shown below.

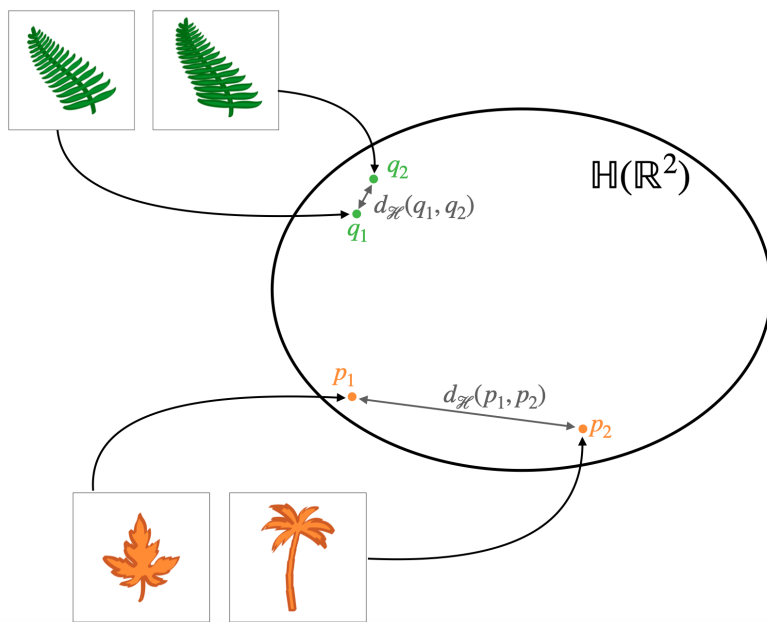


Figure 8: Sets in $\mathbb{H}(\mathbb{R}^2)$ and the distances between them.

Without giving any precise numerical values, it is reasonable to say that the two ferns in green are closer as points in $\mathbb{H}(\mathbb{R}^2)$ with respect to $d_{\mathbb{H}}$ than the leaf and palm tree in orange.

Theorem 2.11 The space $(\mathbb{H}(S), d_{\mathbb{H}})$ is a metric space.

The proof of this theorem can be made easier if we first explore the idea of Hausdorff r -neighborhoods.

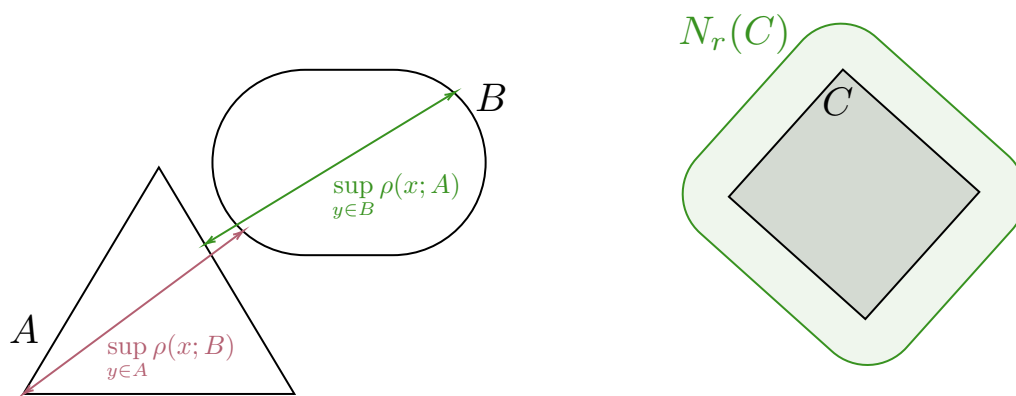


Figure 9: Hausdorff distance (left) and Hausdorff neighborhoods (right).

Definition 2.12 Let $r > 0$. The r -neighborhood of a non-empty set $A \subseteq S$ is the set of points:

$$N_r(A) = \{x \in S \mid \text{there is some } y \in A \text{ with } \rho(x, y) < r\}.$$

Example 2.13 Given a point $x \in S$, the r -neighborhood of **the set** $\{x\}$ is precisely $\mathbb{B}_r(x)$.

Lemma 2.14 Let A, B be points of $\mathbb{H}(S)$, then $d_{\mathbb{H}}(A, B) = \inf\{r > 0 \mid A \subseteq N_r(B) \text{ and } B \subseteq N_r(A)\}$.

Proof of Lemma 2.14. For convenience, label $d_{\mathbb{H}}(A, B) =: d_1$ and $\inf\{r > 0 \mid A \subseteq N_r(B) \text{ and } B \subseteq N_r(A)\} =: d_2$. Let $x \in A$. By definition of the Neighborhood, there exists some $y_0 \in B$ such that $\rho(x, y_0) \leq d_2$. In particular, $\inf_{y \in B} \rho(x, y) \leq \rho(x, y_0) \leq d_2$. This holds for any $x \in A$, so in particular, $\sup_{x \in A} \inf_{y \in B} \rho(x, y) \leq d_2$. Now by symmetry it follows $\sup_{x \in B} \inf_{y \in A} \rho(x, y) \leq d_2$, so $d_1 \leq d_2$. Now we show the reverse inequality. If we show that for any $\epsilon > 0$ one has that $A \subseteq N_{d_1+\epsilon}(B)$ and $B \subseteq N_{d_1+\epsilon}(A)$, then it will follow $d_2 \leq d_1 + \epsilon$ and the claim will follow. Let $x \in A$. By definition of d_1 one has $\inf_{y \in B} \rho(x, y) \leq d_1$. By definition of infimum, there exists some $y_0 \in B$ with $\rho(x, y_0) \leq d_1 + \epsilon$. In particular, $x \in N_{d_1+\epsilon}(B)$. By symmetry one shows the other inclusion and the proof is finished. $\heartsuit$

Proof of Proposition 2.11. Routine checking of the metric axioms:

1. **Symmetry:** follows immediately from the definition.
2. $0 \leq d_{\mathbb{H}}(A, B) < \infty$ **with equality if and only if** $A = B$: Given that we are taking the maximum of two non-negative numbers, non-negativity follows. In addition, $d_{\mathbb{H}}(A, B) = 0$ if and only if $\sup_{x \in A} \rho(x, B) = 0$ and $\sup_{x \in B} \rho(x, A) = 0$. Since A and B are compact, in particular, bounded, it follows that for every $x_a \in A, x_b \in B$, one has $\rho(x_a, B) = \rho(x_b, A) = 0$, and once again, since A and B are closed, it follows $x_a \in B$ and $x_b \in A$. Overall, we have shown that $d_{\mathbb{H}}(A, B) = 0$ if and only if $A \subseteq B$ and $B \subseteq A$. In addition, the fact that A and B are compact implies they are in particular bounded, thus $d_{\mathbb{H}}(A, B) < \infty$.
3. **Triangle Inequality:** this we use the previous lemma. Let $A, B, C \in \mathbb{H}(S)$ and $x \in A$. From the definition of $\inf$ one has that for any $\epsilon > 0$ there exists $y \in B$ such that $\rho(x, y) < d_{\mathbb{H}}(A, B) + \epsilon$. Similarly, given that $y \in B$, there is a $z \in C$ such that $\rho(y, z) < d_{\mathbb{H}}(B, C) + \epsilon$. Combining these two expressions: One has that $\rho(x, y) + \rho(y, z) < d_{\mathbb{H}}(A, B) + d_{\mathbb{H}}(B, C) + 2\epsilon$. We have shown that given any $x \in A$, there exists a $z \in C$ with $\rho(x, z) < d_{\mathbb{H}}(A, B) + d_{\mathbb{H}}(B, C) + 2\epsilon$. This is precisely the definition of $A \subseteq N_{d_{\mathbb{H}}(A, B) + d_{\mathbb{H}}(B, C) + 2\epsilon}(C)$. By symmetry, $C \subseteq N_{d_{\mathbb{H}}(A, B) + d_{\mathbb{H}}(B, C) + 2\epsilon}(A)$. Thus, once again, applying the alternative definition of Hausdorff distance, we see that $d_{\mathbb{H}}(A, C) \leq d_{\mathbb{H}}(A, B) + d_{\mathbb{H}}(B, C) + 2\epsilon$. Given that ϵ was chosen arbitrarily, the triangle inequality follows. $\heartsuit$

Remark 2.15 As one may notice, in the proof of Proposition 2.11, we only used the fact that compact sets are closed and bounded. If we were in $\mathbb{R}^d$, by the Heine-Borel Theorem, that would be equivalent to requiring that our sets were compact. Moving forward, we will keep the more general assumption of our sets being compact, even if strictly speaking it is not needed.

2.3 Other background: Measure Spaces

Most of the mathematical machinery of this project relies on the theory of metric spaces. However, the trail of IFS fractals can also be found in the twin brother of the metric space: the measure space.

Definition 2.16 A measure space is a triplet $(X, \mathcal{F}, \mu)$ where X is a non-empty set, and:

$\mathcal{F}$ is a collection of subsets of X such that:

1. $X \in \mathcal{F}$, $\emptyset \in \mathcal{F}$.
2. $\mathcal{F}$ is closed under countable unions.
3. $\mathcal{F}$ is closed under complements.

$\mu : \mathcal{F} \rightarrow \mathbb{R}$ is a function such that:

1. μ is non-negative.
2. $\mu(\emptyset) = 0$.
3. For a collection (A_n) of pairwise disjoint sets:

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

We call $\mathcal{F}$ a *σ -algebra*, and μ a *measure*. A measure is said to be normalised if $\mu(X) < \infty$. If $\mu(X) = 1$ we refer to the triplet $(X, \mathcal{F}, \mu)$ as a Probability Space.

Measure spaces are to the idea of size what a metric space is to the idea of distances. Sometimes both worlds coalesce and we have a space that is both a metric and a measure space. In this case, there is a special kind of σ -algebra:

Definition 2.17 Let (S, ρ) be a metric space, the *Borel* σ -algebra $\mathcal{B}(S)$ is the smallest (with respect to inclusion) σ -algebra on S that contains the open sets of S . A measure $\mu : \mathcal{B}(S) \rightarrow \mathbb{R}$ is called a *Borel measure*.

One of the main consequences of the ideas of a measure is that it enables us to define a notion of integral that is more general than that of the Riemann Integral, called the Lebesgue Integral. We will assume the reader is familiar with this notion, although we will not use much more than the definition in this project.

3 Iterated Function Systems

3.1 Motivation and mathematical setting

This chapter is dedicated to studying the fractals that arise as fixed points of a certain type of contractive mappings on $\mathbb{H}(S)$.

Definition 3.1 Let (S, ρ) and (P, σ) be metric spaces. A function $f : S \rightarrow P$ is a **contraction** if

$$\sigma(f(x), f(y)) < c\rho(x, y)$$

for any $x, y \in S$ and some $0 < c < 1$, called the **contractivity factor**.

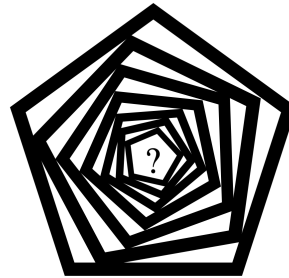


Figure 10: A contraction, what is it tending towards?

The intuition behind this definition is precisely that points in the set get closer after the function is applied to them. For a visual picture of a contraction, see Figure 10, where a contraction is applied to the whole space. It seems this procedure makes the space shrink towards a sort of *black hole*, or a point in the space that *attracts* every other. Our suspicions are justified provided that some conditions on S are met.

Definition 3.2 A metric space (S, ρ) is complete if every Cauchy sequence converges to a point in the space.

Theorem 3.3 (Banach's Fixed Point Theorem) Let (S, ρ) be a complete metric space and $T : S \rightarrow S$ a contraction. There exists a unique point $x \in S$ such that $T(x) = x$ (Called the fixed point).

Definition 3.4 Let $f : S \rightarrow S$ be a map on a metric space. The forward iterates of f , denoted by $f^{\circ(n)}$ are defined to be $f \circ f \circ \dots \circ f$ where this composition occurs n times.

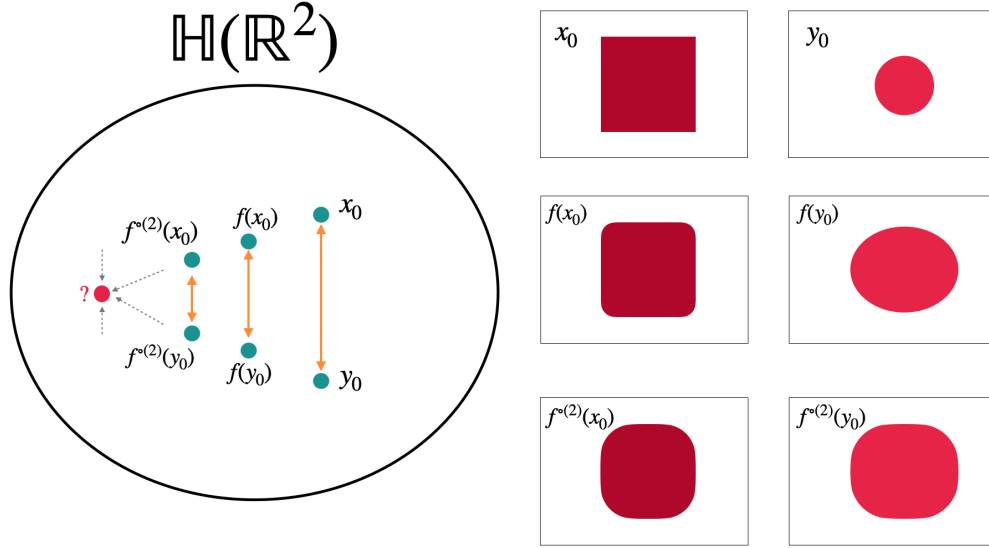
Remark 3.5 This theorem provides not only a theoretical fact but even more importantly provides a way to approximate this fixed point. Namely:

$$x = \lim_{n \rightarrow \infty} T^{\circ(n)}(x_0)$$

where x_0 is **any point** in S !

As hinted in the introduction to the chapter, we will study these maps acting on $\mathbb{H}(S)$ and more importantly, the fixed points they give. Let us consider a heuristic example to understand how contractions act on $\mathbb{H}(S)$:

Example 3.6 Let (S, ρ) be once again regular 2-dimensional Euclidean Space with the usual distance and consider the contraction f on $\mathbb{H}(S)$ depicted on the following diagram. As we can see, if we look at any two points, x_0 and y_0 , they get closer to each other and to the fixed point under the iterated application of f .



We are now ready to state the definition of an Iterated Function System.

Definition 3.7 We refer to a function $f : S \rightarrow S$ as a transformation. A finite collection $\{f_1, \dots, f_m\}$ of continuous transformations is called an **Iterated Function System** (IFS). If each f_i is a contraction, we say $\{f_1, \dots, f_m\}$ is a contractive IFS.

The motivation for this definition is that given several maps from the space S to itself, one can think of “combining them” in order to create a new map that acts on $\mathbb{H}(S)$. There is a very natural way to do this, and it is given by the notion of the Hutchinson operator.

Definition 3.8 Let $\{f_1, \dots, f_m\}$ be an IFS. The **Hutchinson operator** $T : \mathbb{H}(S) \rightarrow \mathbb{H}(S)$ of the IFS is the map defined by:

$$T : A \mapsto \bigcup_{i=1}^m f_i[A].$$

Where $f_i[A]$ is understood to be the set $\{f_i(x) \mid x \in A\}$.

Remark 3.9 Note that by definition, each f_i is continuous, and $T[A]$ is the finite union of continuous images of a compact set. Hence $T[A]$ is compact so T is a well-defined map.

Fractals are very special points in our universe of sets $\mathbb{H}(S)$, and the Hutchinson operator is the boat that will take us close enough to see them.

Example 3.10 Let $S = \mathbb{R}^2$ and $\{f_1, f_2, f_3\}$ be an IFS defined by $f_i(x) = \frac{1}{2}(x - c_i) + c_i$. Where c_i takes the values $(0, 0), (0, 1), (1, 0)$. If we choose:

$$X = \{(a, b) \mid a + b \leq 1, a \geq 0, b \geq 0\}$$

and look at the first few terms of the sequence $(T^{\circ(n)}[X])$ where T is the Hutchinson operator, we can see how there is an intricate self-similar structure being developed (see Figure 11).

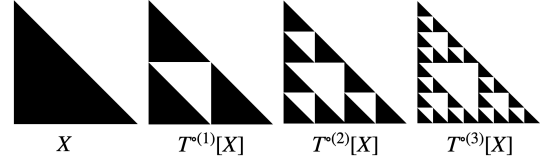


Figure 11: Three iterations of the Hutchinson Operator of an IFS.

3.2 The Attractor Theorem

Referring back to the idea that Hausdorff distance measures *changes in perception*, it is reasonable to believe that in the example above, the elements of the sequence $T^{\circ(n)}[X]$ are getting closer with respect to the Hausdorff distance and that there may be some limit of the sequence. This is indeed the case provided that some conditions are met.

Theorem 3.11 (Attractor Theorem) Suppose (S, ρ) is a complete metric space and $\{f_1, \dots, f_m\}$ a contractive IFS with Hutchinson operator T . There exists a unique non-empty compact set $K \in \mathbb{H}(S)$ such that

$$K = T[K].$$

This set is called the attractor of the IFS.

The choice of the name attractor for the set K in Theorem 3.11 is not by chance, and it rightfully suggests that behind it there will be a mechanism by which sets in $\mathbb{H}(S)$ get attracted (with respect to the Hausdorff metric) towards K . This overall structure is the exact same as Banach's Fixed Point Theorem, so what we expect to find then is that the map T constitutes a contraction on the space $\mathbb{H}(S)$. From now on we will use the words fractal and attractor interchangeably.

3.2.1 Proof of the Attractor Theorem

Now we present all the ingredients for the proof of Theorem 3.11, closely following the discussion of [Edg08, Chapter 2.5]. This will correspond to simply checking the hypotheses of the Fixed Point Theorem.

Proposition 3.12 If (S, ρ) is complete, then the metric space $(\mathbb{H}(S), d_{\mathbb{H}})$ is complete.

Let us sketch the intuition behind the proof of Proposition 3.12 before then proving each step separately: let $(A_n) \subset \mathbb{H}(S)$ be a Cauchy Sequence. We want to show this sequence converges to a point in $\mathbb{H}(S)$, and remarkably, we will do this by constructing its limit without having any other a priori knowledge about (A_n) . The only other information we have about this setup is that S is complete, so we can use the points in each A_k to form sequences in S to help us construct our limit for (A_n) . An educated guess for a possible limit of (A_n) is the set:

$$A := \{x \in S \mid \text{there exists a sequence } (x_k) \text{ with } x_k \in A_k \text{ and } x_k \rightarrow x\}.$$

Lemma 3.13 $d_{\mathbb{H}}(A_n, A) \rightarrow 0$.

Proof. The heuristics are the following: First let $\epsilon > 0$ and by choosing n to be large enough, let a point $y \in A_n$ be given. Using the fact that (A_k) is Cauchy, we will construct a Cauchy sequence $(x_k) \subseteq S$ related to our initial point y by taking each x_k in a smart way from each A_k . By completeness of S , this sequence (x_k) will converge to some point x . This is the precise meaning of x being in A as stated above. We will have constructed the sequence in such a way that $\rho(x, y) < \epsilon$ and as such, one has $A_n \subset N_{\epsilon}(A)$. Then we will show the converse inclusion.

In light of the above, choose N large enough such that whenever $n, m \geq N$, one has $d_{\mathbb{H}}(A_n, A_m) < \epsilon$. Let $y \in A_n$ be given. Recall that we aim to show there exists an $x \in A$ such that $\rho(y, x) < \epsilon$. We will start by creating an increasing sequence of natural numbers:

$$N = k_1 < k_2 < \dots$$

Such that whenever $n, m \geq k_j$ one has: $d_{\mathbb{H}}(A_n, A_m) < 2^{-j}\epsilon$. In particular, whenever $n \geq k_j$, one has that $d_{\mathbb{H}}(A_n, A_{k_j}) < 2^{-j}\epsilon$. Note that for any $n > N$, one can find a k_j such that $k_j < n < k_{j+1}$, and in particular: given any $y_{k_j} \in A_{k_j}$ there will be some $y_n \in A_n$ such that $\rho(y_{k_j}, y_n) < 2^{-j}\epsilon$. This justifies the creation of the following sequence of points in S :

$$y_n = \begin{cases} \text{any } y \in A_n \text{ whenever } n \leq N. \\ \text{some } y \in A_n \text{ such that } \rho(y, y_{k_j}) < 2^{-j}\epsilon \text{ where } k_j < n < k_{j+1} \text{ whenever } n > N. \end{cases}$$

Let us check this sequence is indeed Cauchy: for any $m \geq n \geq N$, we see that by construction there are some $k_j < k_{j+1} < \dots < k_l < k_{l+1}$ such that:

$$\rho(y_m, y_n) \leq \rho(y_n, y_{k_j}) + \rho(y_{k_j}, y_{k_{j+1}}) + \dots + \rho(y_{k_l}, y_m) < \sum_{t=j}^l 2^{-t}\epsilon < \epsilon.$$

By hypothesis, S is complete so there is a limit $x \in S$ to this sequence. As illustrated in the Heuristics, this means $x \in A$. Now it is easy to check that $\rho(y, x) < \epsilon$. Indeed: we know there is some N large enough so that: $\rho(y_N, x) < \epsilon/2$ and moreover, due to the way (y_n) was constructed, we see that for any y_k , we have $\rho(y, y_k) < \epsilon/2$, so we can see that: $\rho(y, x) < \rho(y, y_N) + \rho(y_N, x) = \epsilon$.

Conversely, suppose $x \in A$. Our goal is now to show that for large enough n , there is some $y \in A_n$ such that $\rho(x, y) < \epsilon$. By assumption, let (x_n) be a sequence with $x_k \in A_k$ and $x_n \rightarrow x$. Given any $\epsilon > 0$ we know that for k large enough, $\rho(x_k, x) < \frac{\epsilon}{2}$. If $k, n > N$, one has that $d_{\mathbb{H}}(A_k, A_n) < \frac{\epsilon}{2}$, and thus there exists some $y \in A_n$ such that $\rho(x_k, y) < \frac{\epsilon}{2}$. Thus $\rho(x, y) \leq \rho(x, x_k) + \rho(x_k, y) = \epsilon$ as required. $\heartsuit$

Notice that in this proof, we have already shown that the set A is not empty. Therefore all left to show is that the set A is compact. To do so we need the following lemmas:

Lemma 3.14 The set A is totally bounded.

Proof. Let $\epsilon > 0$ be given and choose $N \in \mathbb{N}$ large enough such that $d_{\mathbb{H}}(A_N, A) < \epsilon/3$. Recall that $A_N \in \mathbb{H}(S)$, thus A_N is compact and in particular totally bounded. Let $\{y_1, \dots, y_m\}$ be an $\epsilon/3$ net of A_N . Noting that $A_N \subseteq N_{\epsilon/3}(A)$, one has that for each y_i , there exists an $x_i \in A$ such that $\rho(x_i, y_i) < \epsilon/3$. The claim is that the set $B = \{x_i \mid i = 1, \dots, m\}$ is an ϵ net of A .

Indeed, let $x \in A$. Noting that $A \subseteq N_{\epsilon/3}(A_N)$, we see that there is a $y \in A_N$ such that $\rho(x, y) < \epsilon/3$. In addition, having that $y \in A_N$, we can use the net constructed on A_N to choose an y_j such that $\rho(y, y_j) < \epsilon/3$. Moreover, by the argument on the last paragraph, there is an x_j from the set B of such that $\rho(y_j, x_j) < \epsilon/3$. Putting these three things together: $\rho(x, x_j) < \rho(x, y) + \rho(y, y_j) + \rho(y_j, x_j) < \epsilon$. As claimed. $\heartsuit$

Lemma 3.15 The set A is closed.

Proof. Let $\epsilon > 0$ be given and $x \in \bar{A}$ be an element in the closure of A , we will show $x \in A$. Let $(y_n)_{n \in \mathbb{N}}$ be a sequence in A that converges to x . We can choose N large enough so that both $\rho(x, y_n) < \epsilon/2$ and $d_{\mathbb{H}}(A_n, A) < \epsilon/2$ whenever $n \geq N$. Noting that $A \subseteq N_{\epsilon/2}(A_n)$, we have that there exists some $z_n \in A_n$ such that $\rho(z_n, y_n) < \epsilon/2$. Combining these results, we have that $\rho(x, z_n) \leq \rho(x, y_n) + \rho(y_n, z_n) < \epsilon$. It follows $z_n \rightarrow x$. This satisfies the definition of x belonging to the set A . Thus finishing the proof. $\heartsuit$

Lemma 3.16 The set A is compact.

Proof. We will show A is sequentially compact. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in A . We show (x_n) has a subsequence (y_n) converging to a point $y \in A$.

Let $F = \{x_n \mid n \in \mathbb{N}\}$. If F is finite, then (x_n) only takes finitely many distinct values so the proof is done. Hence, assume F is infinite. A is totally bounded so let $\mathcal{B}_1$ be a $1/2$ -net of A . By assumption on F , there is some point $z_1 \in \mathcal{B}_1$ such that the ball of radius $1/2$ about z_1 , $\mathbb{B}_{\frac{1}{2}}(z_1)$ contains infinitely

many points of F . Call the set of such points F_1 .

Subsets of totally bounded spaces are in themselves totally bounded so let $\mathcal{B}_2$ be a $1/4$ -net of F_1 . By the same argument, there must be some $z_2 \in \mathcal{B}_2$ such that $\mathbb{B}_{\frac{1}{4}}(z_2)$ contains infinitely many points of F . Call the set of such points F_2 .

Repeating this we obtain a decreasing sequence of sets F_j , with $\text{diam}(F_j) < 2^{-j}$. Choose the subsequence (y_n) of (x_n) by letting $y_n \in F_n$. Observe that if $j < k$ then $\rho(y_j, y_k) < 2^{-j}$, hence (y_n) is Cauchy. By assumption of S being complete, we have $y_n \rightarrow y$. By A being closed, we see that $y \in A$. Thus finishing the proof. $\heartsuit$

Combining the previous lemmas yields a proof for Proposition 3.12. Now there is only one ingredient left to prove the Attractor Theorem, namely the second hypothesis for the Fixed Point Theorem.

Lemma 3.17 The Hutchinson operator $T : \mathbb{H}(S) \rightarrow \mathbb{H}(S)$ of a contractive IFS $\{f_1, \dots, f_m\}$ is a contraction.

Proof. Each f_i being a contraction means that $\rho(f_i(x), f_i(y)) < r_i \rho(x, y)$ for all $x, y \in S$. Let r be the maximum among all the r_i with $i = 1, \dots, m$. The claim is that $d_{\mathbb{H}}(T[A], T[B]) < r d_{\mathbb{H}}(A, B)$ for any $A, B \in \mathbb{H}(S)$.

Let $\epsilon > 0$ be given and let $q = d_{\mathbb{H}}(A, B) + \frac{\epsilon}{r}$. We prove the claim by showing $F[A] \subseteq N_{rq}(F[B])$ and vice-versa. Indeed, let $x \in F[A]$. There must be some $x' \in A$ and some $1 \leq i \leq m$ such that $x = f_i(x')$. Note that $A \subseteq N_q(B)$ so given that $x' \in A$, we can choose some $y' \in B$ with $\rho(x', y') < q$. Let $y = f_i(y')$. We see that $\rho(x, y) = \rho(f_i(x'), f_i(y')) < r_i \rho(x', y') < r q + \epsilon$. Recall x was chosen arbitrarily so $F[A] \subseteq N_{rq+\epsilon}(F[B])$. By symmetry one can show the reverse inclusion.

In conclusion, one sees that $d_{\mathbb{H}}(F[A], F[B]) < r q + \epsilon < r d_{\mathbb{H}}(A, B) + \epsilon$. Thus the claim is proven. $\heartsuit$

Remark 3.18 Not only does the Attractor Theorem guarantee the existence of IFS fractals, but also gives us a way of approximating them by a sequence of sets: if K is the Attractor of a contractive IFS, i.e: $K = T[K]$, then:

$$K = \lim_{n \rightarrow \infty} T^{\circ(n)}[K_0]$$

For any $K_0 \in \mathbb{H}(S)$.

After having hopefully convinced the reader about the existence (and uniqueness!) of the attractors of contractive IFS, we now get to the fun part, which is to generate images of them! As hinted in the discussion of section 1.2, if we want to Ray-March IFS fractals, we will have to estimate the distance to them. There are several methods for this, and although not all are suitable to Ray-March IFS fractals, we will now give an overview of them. From now on, we will assume (S, ρ) to be a complete Metric Space.

4 Distance Estimation Problem: Methods in the Literature

The methods in the literature that can be used to solve the distance estimation problem can be broadly classified into the following classes: on the one hand, we have methods that aim to obtain a finite sample of points in the fractal and then estimate the distance function as the minimum distance to each of these points. On the other hand, we have methods that using some deeper notions of IFS Theory, study the orbits of points under the iteration of the maps that constitute the IFS. We will now give an overview of these methods. For the rest of this discussion, let $\{f_1, \dots, f_m\}$ be contractive IFS, with each f_i having a contractivity factor of r_i . Let T be the Hutchinson operator and $K = T[K]$ be the attractor of the IFS.

4.1 Sampling Methods

The first two methods that we present were introduced in [HPS90], and constituted the first attempt at producing an algorithm that takes in an arbitrary contractive IFS and outputs an image of the attractor. Even though they have seen improvement since first introduced (see [Mar03], [Mar01]), we will present these algorithms in their most rudimentary form to showcase only the fundamental ideas.

4.1.1 Fixed Approximation

This is in some sense the naive approach to distance estimation, and it is also named the *bounding volumes* method for reasons that will become clear soon. The goal is to construct a finite set of points K_ϵ that approximates K , i.e: $d_{\mathbb{H}}(K_\epsilon, K) < \epsilon$ for a given tolerance $\epsilon > 0$. Once this set is constructed, we use the fact that K_ϵ is finite alongside the following result to finish our task of estimating the distance:

Proposition 4.1 Let $A, B \in \mathbb{H}(S)$. If $d_{\mathbb{H}}(A, B) < \epsilon$, then for any $x \in S$, one has $|\rho(x; A) - \rho(x; B)| < \epsilon$.

Proof. Since $A, B \in \mathbb{H}(S)$, there exists some y_A, y_B in A, B respectively such that for any given $x \in S$, $\rho(x; A) = \rho(x, y_A)$ and $\rho(x; B) = \rho(x, y_B)$. Since $d_{\mathbb{H}}(A, B) < \epsilon$, by Lemma 2.14, there exists some $z_A \in A, z_B \in B$ such that $\rho(y_A, z_B) < \epsilon$ and $\rho(y_B, z_A) < \epsilon$. We now combine this to see that:

$$\rho(x; B) \leq \rho(x, z_B) < \rho(x, y_A) + \epsilon = \rho(x; A) + \epsilon.$$

Similarly, we can show $\rho(x; A) < \rho(x; B) + \epsilon$. Thus finishing the claim. ♡

We now turn our attention to construct the set K_ϵ . Thankfully the structure of the IFS itself provides a natural way to do this. We summarise this method in the following proposition. A similar, less detailed justification for the validity of the algorithm can be found in [HD91, Section 2.3].

Proposition 4.2 Let $x_0 \in K$ be any point in the attractor. Consider the set of finite index sequences:

$$I = \left\{ (k_1, k_2, \dots, k_n) \mid 1 \leq k_i \leq m \text{ with } r_{k_1} r_{k_2} \dots r_{k_n} < \frac{\epsilon}{\text{diam}(K)} \right\}$$

Then the set

$$K_\epsilon = \{x \in S \mid x = f_{k_1} \circ f_{k_2} \circ \dots \circ f_{k_n}(x_0), (k_1, \dots, k_n) \in I\}$$

has the property that $d_{\mathbb{H}}(K, K_\epsilon) < \epsilon$.

Proof. First, we note that $K_\epsilon \subseteq K$. Thus for any $\delta > 0$, we have $K_\epsilon \subseteq N_\delta(K)$. Therefore all we need to show is that $K \subseteq N_\epsilon(K_\epsilon)$, or in other words, that given any $x \in K$, we can find an $x' \in K_\epsilon$ such that $\rho(x, x') < \epsilon$. For convenience, let $d = \text{diam}(K)$.

Let us first show that $f_i[K] \subseteq \mathbb{B}_{r_i d}(f_i(x_0))$. Indeed, given any $x \in f_i[K]$ we can express $x = f_i(x')$ for some $x' \in K$ and as such:

$$\rho(x, f_i(x_0)) = \rho(f_i(x'), f_i(x_0)) \leq r_i \rho(x', x_0) \leq r_i d.$$

Since $K = \bigcup_{i=1}^m f_i[K]$, one has that $K \subseteq \bigcup_{i=1}^m \mathbb{B}_{r_i d}(f_i(x_0))$. Moreover, since each $r_i < 1$, one may iterate this step and eventually one will have that:

$$K \subseteq \bigcup_{k_1=1}^m \bigcup_{k_2=1}^m \dots \bigcup_{k_n=1}^m \mathbb{B}_{r_{k_1} \dots r_{k_n} d}(f_{k_1} \circ \dots \circ f_{k_n}(x_0))$$

With $r_{k_1} \dots r_{k_n} d < \epsilon$. Notice that each of these balls is centered at a point belonging to K_ϵ . In conclusion, we have shown that given any $x \in K$, there is some $x' = f_{k_1} \circ \dots \circ f_{k_n}(x_0)$ such that $\rho(x, x') < \epsilon$. Thus finishing the claim that $d_{\mathbb{H}}(K, K_\epsilon) < \epsilon$.

♡

The set I can be found through a tree traversal algorithm and thus one can algorithmically find K_ϵ . Pseudo-code for this can be found in [HPS90, Page 10]. The reason why this is a naive approach is that the time and memory required to compute this estimation is directly proportional to $|K_\epsilon|$, which in itself depends on how small ϵ is. However, the key observation is that to find an approximation to the distance between a point and K , one does not need to uniformly approximate K , but rather only a small region of the set. This is exploited in the following algorithm.

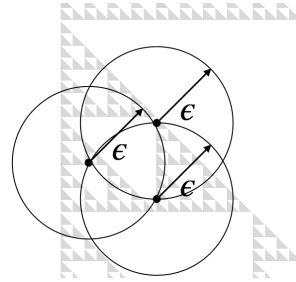
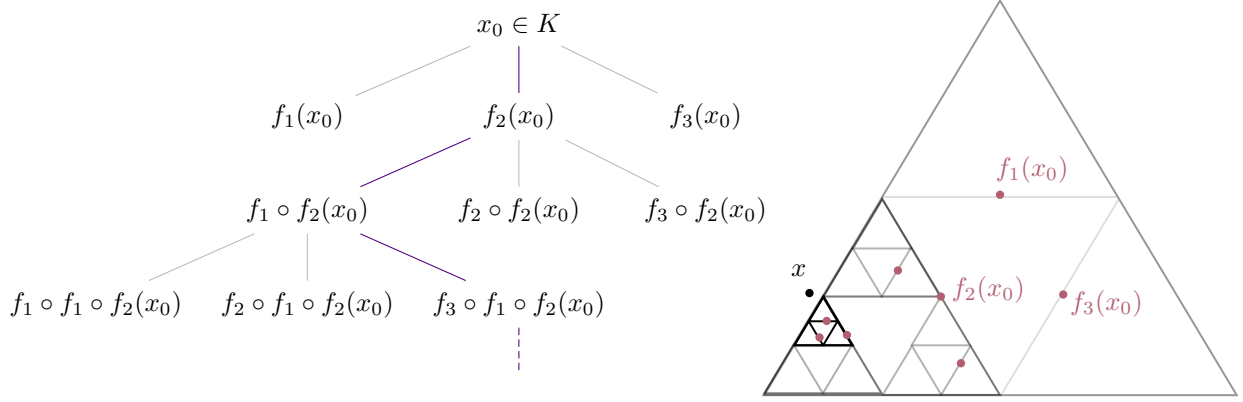


Figure 12: An illustration of the Fixed Approximation method, some points of K_ϵ have been shaded.

4.1.2 Region Subdivision: an improvement to Fixed Approximation

This algorithm, introduced in [HPS90, Section 3.2] under the name of *Adaptive Cut Algorithm*, can be seen as the successor of the Fixed Approximation: it is very similar in philosophy, but improves the performance of its predecessor by cutting the number of points to check. This is done by applying the simple observation that the Fixed Approximation method partitions K into smaller regions: $f_i[K]$ for $i = 1, \dots, m$ and in turn partitions each of these smaller regions. However, out of all these smaller regions, only one, $f_{\mu_1}[K]$, minimises the distance to our point. Therefore when continuing the search for points in K_ϵ , the points that belong to the regions $f_i[K]$ where $i \neq \mu_i$ can all be discarded, thus saving computation time and a substantial amount of memory. We will not provide proof of the validity of this algorithm, since we shall prove a very similar result in section 6.1. However, for an intuitive illustration, we can see the tree diagram below, where the shaded line indicates the path that minimises distance at each layer of the search.



Pseudo-code of the Region Subdivision algorithm can be found in [HPS90, Page 18]. In summary: the first two methods presented exploited the construction of the IFS to obtain a “sample” of the fractal, and then approximated the distance to this sample. As we saw in the proof of Proposition 4.2, this corresponds to covering the attractor by spheres of decreasing radius. Unsurprisingly, Ray–Marching the first few iterations reflects this:

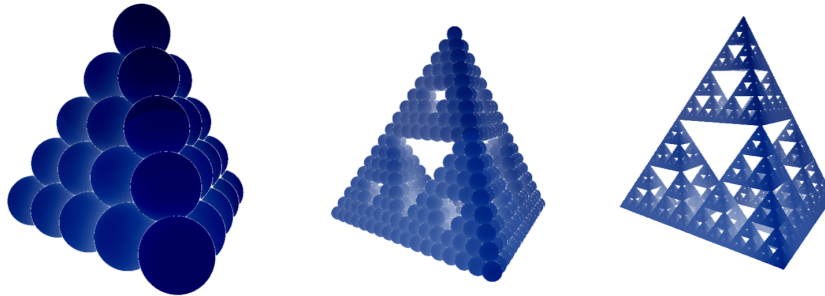


Figure 13: Region Subdivision method. Images were obtained through the code of Inigo Quilez [Quic].

4.2 Escape Time Algorithm

This method tackles fractal visualisation through the lens of dynamical systems. In particular, by considering the dynamics of *the inverse* of the IFS. The notion that makes this precise is that of *shift dynamical systems*. The following discussion will serve as a motivation for the escape time algorithm and a surface-level introduction to the notions of chaotic dynamics on fractals. We follow the discussion of [Bar93, Chapter 4, Chapter 7].

Definition 4.3 Let $\{f_1, \dots, f_m\}$ be a contractive IFS with attractor K . We say the IFS is **totally disconnected** if $f_i[K] \cap f_j[K] = \emptyset$ for all $i \neq j$.

Remark 4.4 It can be shown [Bar93, Theorem 2.1] that this definition is equivalent to saying that every point $y \in K$ can be reached from starting at any point $x \in K$ by applying a **unique** sequence of the transformations $\{f_1, \dots, f_m\}$ to x . This result should come as natural to the reader as a simple consequence of the non-overlapping condition of Definition

This definition justifies the construction of the *shift transformation* associated with an IFS, which as we will see, can be thought of as “the inverse” of an IFS, and will motivate the introduction of the escape time algorithm for fractal visualisation.

Definition 4.5 Let $\{f_1, \dots, f_m\}$ be a contractive, invertible, totally disconnected IFS with attractor K . The associated **shift transformation** on K is the transformation $F : K \rightarrow K$ given by:

$$F : x \mapsto f_i^{-1}(x) \quad \text{for } x \in f_i[K]$$

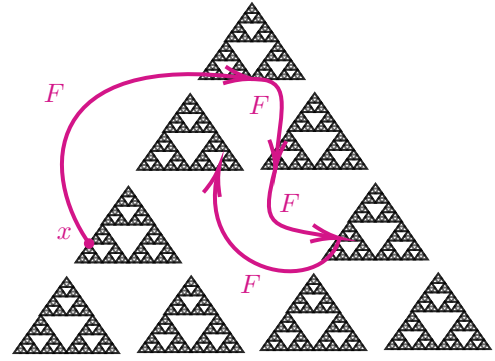


Figure 14: A shift transformation on a totally disconnected IFS.

Proposition 4.6 Let F and K be as in Definition 4.5. Then $F[K] = K$.

Proof. Indeed: Given any $x \in K$, since we know that K is covered by $f_j[K]$ for $j = 1 \dots m$, it follows $x = f_{j'}(x')$ for some j' and $x' \in K$. Thus $F(x) = f_{j'}^{-1}(f_{j'}(x')) = x' \in K$. Thus establishing that $F[K] \subseteq K$. Conversely, let $x_0 \in K$ be given. By a similar argument, $x_0 = f_{j'}(x'_0)$ for some j' , x'_0 . Therefore, $S(f_{j'}(x_0)) = f_{j'}^{-1}(f_{j'}(x_0)) = x_0$. Showing that $K \subseteq S[K]$. $\heartsuit$

Proposition 4.7 (Exercise 1.7, Chapter 7, [Bar93]) Let $\{f_1, \dots, f_m\}$ be a contractive, totally disconnected IFS with attractor K in $\mathbb{R}^n$ and let its shift transformation be $F : K \rightarrow K$. Then F extends to a transformation $F' : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that when restricted to K : $F'|_K = F$.

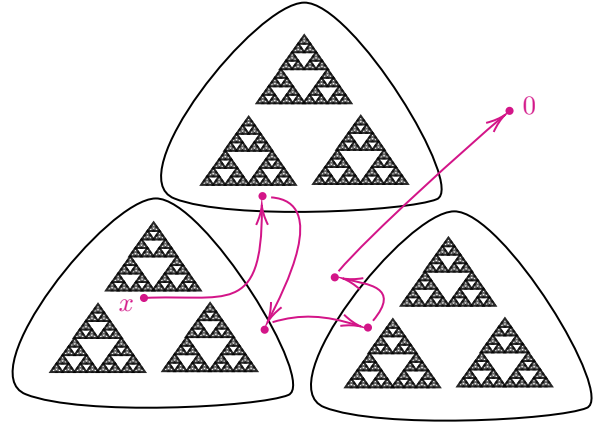
Proof. Recall that K is compact. Given that the IFS is contractive, each map f_i is continuous so each $f_i[K]$ is also compact. Since the IFS is totally disconnected, we have that $f_i[K] \cap f_j[K] = \emptyset$. Therefore, for any $i \neq j$, we have $d_{\mathbb{H}}(f_i[K], f_j[K]) > 0$. Let $\epsilon = \min_{i \neq j} d_{\mathbb{H}}(f_i[K], f_j[K])$ and define the following functions:

$$f'_i(x) = \begin{cases} \left(1 - \frac{\rho(x; f_i[K])}{\epsilon}\right) f_i^{-1}(x) & \text{if } \rho(x; f_i[K]) \leq \epsilon/2 \\ 0 & \text{otherwise.} \end{cases}$$

Finally, we define $F'(x) = \sum_{i=1}^m f'_i(x)$. Notice that if $x \in K$, then all but one of the summands are zero, and the non-zero summand is precisely $f_i^{-1}(x)$ for some i . Therefore, F' is an extension of F as required. $\heartsuit$

Remark 4.8 (Interpretation of the extension)

Without loss of generality, we may assume that the point 0 (which we shall refer to as the origin) is far away enough from K , so that we may interpret reaching the origin as having “escaped the fractal”. See the diagram to the right for a heuristic illustration of this process.



Proposition 4.9 Let $\{f_1, \dots, f_m\}, F'$, and K be as in Proposition 4.7. For any starting point $x_0 \in \mathbb{R}^n \setminus K$, the orbit $\{F'^{\circ(n)}(x_0) \mid n \geq 0\}$ eventually escapes the fractal.

Proof. Recall that we defined ϵ to be $\min_{i \neq j} d_{\mathbb{H}}(f_i[K], f_j[K])$ so notice that it is enough to show that the distance to K eventually is greater than $\epsilon/2$, indeed, if some point x_n in the orbit of x_0 under the iterated application of F' has that $\rho(x_n; K) > \epsilon/2$, in particular it follows that for each $i = 1, \dots, m$, $\rho(x_n; f_i[K]) > \epsilon/2$ and as such, all the individual maps f'_i collapse to zero and as such $F'(x_n) = 0$.

If the starting distance of x_0 to K is already greater than $\epsilon/2$, there is nothing to prove, so suppose $\rho(x_0; K) \leq \epsilon/2$. In particular, it follows that $x_0 \in N_{\epsilon/2}(f_j[K])$ for some unique $j \in \{1, \dots, m\}$. Thus

$$\rho(F'(x_0); K) = \rho\left(\left(1 - \frac{\rho(x_0; f_j[K])}{\epsilon}\right) f_j^{-1}(x_0); f_j[K]\right)$$

The origin can be chosen to be far away enough from K so that this quantity is greater than or equal to $\rho(f_j^{-1}(x_0); f_j[K])$. Notice that $f_j[K]$ is compact so there is some $y_0 \in K$ such that $\rho(f_j^{-1}(x_0); f_j[K])$ is equal to $\rho(f_j^{-1}(x_0), y_0)$. We can write this last quantity as $\rho(f_j^{-1}(x_0), f_j^{-1} \circ f_j(y_0))$. Since f_j is a contraction with contractivity factor r_j , this last quantity is greater than or equal to $r_j^{-1} \rho(x_0; f_j(y_0))$, noticing that $f_j(y_0) \in f_j[K]$, we see that $r_j^{-1} \rho(x_0; f_j(y_0)) \geq r_j^{-1} \rho(x_0; f_j[K]) = r_j^{-1} \rho(x_0; K)$. Noting that r_j^{-1} is strictly greater than 1, we deduce that after repeatedly applying F' , one will have that the distance to K will surpass $\epsilon/2$ as required. $\heartsuit$

We have now learned that considering inverses of an IFS will give us interesting behaviour, in particular, that the attractor, whilst still being an invariant set under these new dynamics, has now turned into a *repeller* that makes points outside it eventually escape under iteration of the shift transformation. The question that we are interested in studying now is: what information can we obtain by studying *how quickly* points escape the fractal under iteration of its inverse maps? Let us illustrate this with the following famous example:

Example 4.10 (Julia sets) Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be the polynomial $f(z) = z^2 + c$ where $c \in \mathbb{C}$ is fixed. The filled-in Julia set is:

$$K_f = \{z \in \mathbb{C} \mid \{f^{o(n)}(z) \mid n \geq 0\} \text{ is bounded}\}$$

Then the Julia set J_f is defined as the boundary of K_f . It can be shown [Bea91, Theorem 3.2.4] that $J_f = f[J_f] = f^{-1}[J_f]$, so J_f can be thought as the fixed point of the IFS $\{-\sqrt{z-c}, \sqrt{z+c}\}$ (Notice that this is not a contractive IFS in the strict sense we have defined it, so the results we have shown will not apply here. Nonetheless we just use this example as an illustration). The shift transformation of this IFS is precisely the map polynomial f , since both maps of the IFS collapse to the same inverse.

We can use a straightforward algorithm to visualise the set: fix some “escape radius” $R > 0$ and some maximum number of iterations $N > 0$. Compute $\{z, f(z), \dots, f^{o(N)}(z)\}$ and colour the pixel “on coordinate z ” if and only if the orbit remains within distance R of the origin. This gives very rudimentary images, but if we modify the algorithm to take into account how many iterations the orbit takes to escape, we can produce images as shown in Figure 15. It appears that the colour scheme follows a rule based on the distance to the fractal: the closer it is, the longer it takes to escape and vice-versa. As it turns out, one can formalise these intuitions and show that there is a connection between distance to the attractor and the escape time.

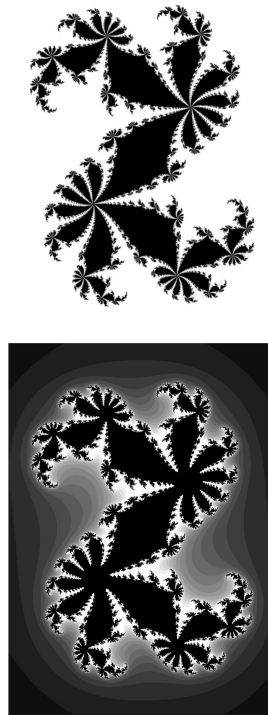


Figure 15: A filled-in Julia set K_f for $c = 0.36 + 0.1i$ coloured using the “rudimentary” and “escape-time” colourings.

Definition 4.11 Let $\{1, \dots, m\}$ be a contractive, invertible IFS with attractor K and let $y \in K$ be fixed. Let $R > 0$ sufficiently large such that $f_i[\mathbb{B}_R(y)] \subseteq \mathbb{B}_R(y)$ for all $i = 1, \dots, m$. Then the escape-time function $E_R : S \setminus K \rightarrow \mathbb{N} \cup \{0\}$ is defined as:

$$x \mapsto \begin{cases} 0 & \text{if } \rho(x, y) \geq R \\ \max_{1 \leq i \leq m} \frac{\log(R) - \log(\rho(x, y))}{\log(\rho(f_i^{-1}(x)) - \log(\rho(x, y)))} & \text{if } \rho(x, y) < R \text{ and } \rho(f_i^{-1}(x), y) \geq R \text{ for } i = 1, \dots, m \\ 1 + \max_{1 \leq i \leq m} E_R(f_i^{-1}(x)) & \text{otherwise} \end{cases}$$

The middle condition, which is triggered only when the point is “about to escape”, should be thought of as a technicality that can be used to show [HPS90, Theorem 4] that the escape-time function is continuous both in R when fixing x , and x when fixing R . For a more natural intuition, one can remove the middle condition, which gives the so-called integer escape-time function. Notice that the integer escape time function represents the same escape-time as what was discussed in the case of Julia sets before. We will now state without proof a result that unifies this discussion back to our goal of studying the distance estimation problem.

Theorem 4.12 ([HPS90], Theorem 8) .

Let $\{f_1, \dots, f_m\} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an IFS with attractor K , consisting of affine invertible contractions such that each transformation has the same contracting ratio $r < 1$. Then the **limit escape-time function**, defined as:

$$E_\infty(x) : \mathbb{R}^n \setminus K \rightarrow \mathbb{R} \quad x \mapsto \lim_{R \rightarrow \infty} \left(E_R(x) + \frac{\log(R)}{\log(r)} \right)$$

Is the logarithm base r of the distance function:

$$E_\infty(x) = \log_r(\rho(x; K))$$

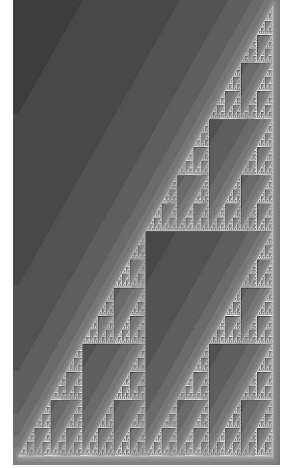


Figure 16: Visualisation of the Sierpinski Triangle. Pixels are coloured based on the integer escape-time of the shift transformation.

In his thesis, Hepting used this method to produce 3-D images of fractals by interpreting the escape-time as height values [Hep91, Figure 5.13], and although this does not technically constitute Ray-Marching, it shows this method can be used for 3-D visualisation of fractals. Unfortunately, in order for us to use this algorithm in a justified manner to Ray-March fractals, we would have to restrict ourselves to only affine contractions. We would like to explore more general methods so we will not consider this technique any further. This is not to say that this method cannot produce interestingly looking fractals. For some impressive artwork that uses this principle, see [kni].

4.3 Invariant Measures

The final method we present, even though not used for Ray-Marching fractals, also originates in visualisation purposes and offers an interesting solution to the distance estimation problem. Along the way we will see some important results in the Theory of Iterated Function Systems, amongst them the existence of measures supported on attractors.

Definition 4.13 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. The support of μ is the set of points:

$$\{x \in \Omega \mid \forall \epsilon > 0, \mu(\mathbb{B}_\epsilon(x)) > 0\}.$$

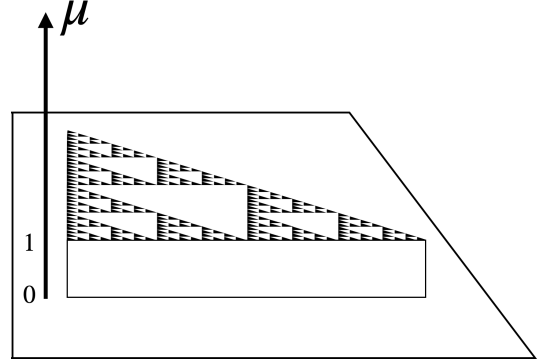


Figure 17: Illustration of a measure supported on a fractal.

We will use invariant measures to solve the distance estimation problem: as we will prove, the distance from a point x to the fractal will be given by the radius of the largest ball centered at x that has measure zero. The existence of measures supported on fractals should be thought of as the twin result of Theorem 3.11 due to the similarity of both proofs. Giving thorough proof of the facts that are about to be presented is outside the scope of this project, however, we will give an overview of some main results which will be crucial to solve the distance estimation problem. It all begins by considering the following set:

Definition 4.14 Let (S, ρ) be a metric space and let $\mathcal{B}$ be its Borel σ -algebra. The set $\mathbb{P}(S)$ is defined to be the set of all its normalised Borel measures:

$$\mathbb{P}(S) = \{\mu : \mathcal{B} \rightarrow [0, \infty) \text{ such that } \mu(S) = 1\}.$$

Much like how a clever function was chosen to make $\mathbb{H}(S)$ a Metric Space, one can equip $\mathbb{P}(S)$ with the following function:

Definition 4.15 Let $\mu, \nu \in \mathbb{P}(S)$, the **Hutchinson distance** between μ and ν , is defined to be:

$$d_{\mathbb{P}}(\mu, \nu) = \sup \left\{ \left| \int_S f d\mu - \int_S f d\nu \right| \mid f \in \text{Lip}_1(S) \right\},$$

where $\text{Lip}_1(S)$ is the set of Lipschitz functions from S to $\mathbb{R}$ with Lipschitz constant 1.

Which can be shown to turn $(\mathbb{P}(S), d_{\mathbb{P}})$ into a metric space [Hut81, Section 4.3], and that if (S, ρ) is compact then so is $(\mathbb{P}(S), d_{\mathbb{P}})$ (see [Bar93, Page 350], for a sketch of proof). As a result $(\mathbb{P}(S), d_{\mathbb{P}})$ is complete.

Recall how in Section 3, we constructed the Hutchinson operator by considering how an IFS could act on sets of $\mathbb{H}(S)$. Using this same inspiration, one can construct an analogous operator that acts on $\mathbb{P}(S)$:

$$M : \mathbb{P}(S) \rightarrow \mathbb{P}(S) \quad M : \mu \mapsto \frac{1}{m} \sum_{i=1}^m \mu \circ f_i^{-1}$$

This operator is well defined provided the maps f_i are measurable, however in our context each f_i is a contraction, hence continuous. Therefore this definition is justified. Moreover, just as T was a contraction mapping with respect to $d_{\mathbb{H}}$, it turns out [Hut81, Section 4.4] that M is a contraction mapping with respect to $d_{\mathbb{P}}$! Hence there exists a unique probability measure $\mu \in \mathbb{P}(S)$ such that $M(\mu) = \mu$. This is called the **invariant measure** of the IFS.

Observe what has happened here: by considering $(\mathbb{H}(S), d_{\mathbb{H}})$ and $(\mathbb{P}(S), d_{\mathbb{P}})$ - two fundamentally different spaces, one can show the existence of two special objects, each native to their corresponding space, and both invariant under the “action of an IFS”. It almost feels like a tragic romance: forever connected but separated by different universes. Thankfully, this love story has a happy ending, and it comes in the shape of the following theorem, which forever unites both results:

Theorem 4.16 ([Bar93], Theorem 6.2) Let μ and K be the invariant measure and attractor of a contractive IFS. Then the support of μ is K .

We are now ready to apply this theory back to the visualisation of fractals. Barnsley [Bar93, Chapter 9, Section 8] suggests a method to visualise a fractal by plotting the invariant measure on the screen. The algorithm assigns an open ball $\mathbb{B}_{\epsilon}(P_{ij})$ of a constant prescribed radius to each pixel P_{ij} on the screen. Then one proceeds to compute the measure $\mu(\mathbb{B}_{\epsilon}(P_{ij}))$ and colour the pixel accordingly. To compute this measure, one can use the following theorem due to Elton:

Theorem 4.17 ([Elt87], Corollary 2) Let $\{f_1, \dots, f_m\}$, μ and K be the same as in Theorem 4.16 and let B a Borel set such that $\mu(\partial B) = 0$. Define the following sequence:

$$x_n = f_{\sigma_n}(x_{n-1}) \quad \text{with } x_0 \in K \text{ any point in the attractor}$$

where each σ_n is an IID random variable with distribution $\text{Unif}(\{1, 2, \dots, m\})$. Then:

$$\mu(B) = \lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{k=0}^n \delta_{\{x_k\}}(B).$$

In simple words, given a point $x_0 \in K$, one can approximate the measure of a Borel set B whose boundary has measure zero by computing a large enough number of elements of the orbit $\{x_0, f_{\sigma_1}(x_0), f_{\sigma_2}(f_{\sigma_1}(x_0)), \dots\}$ and counting how many of those elements are located inside B . One can exploit this method to estimate the distance function of an attractor K whose invariant measure is μ :

Proposition 4.18 Given a point $x \in S$ one has $\rho(x; K) = \sup\{r > 0 \mid \mu(\mathbb{B}_r(x)) = 0\}$.

Proof. Let $S = \sup\{r > 0 \mid \mu(\mathbb{B}_r(x)) = 0\}$. We will show $\rho(x; K) \leq S$ and $\rho(x; K) \geq S$.

1. $S \leq \rho(x; K)$: Suppose to the contrary that $S > \rho(x; K)$. I.e: there exists some $\delta > 0$ such that $S = \rho(x; K) + \delta$. On the one hand, since K is compact, there exists some $k \in K$ such that $\rho(x; K) = \rho(x, k)$. On the other hand, one has $\mu(\mathbb{B}_{\rho(x, k) + \delta/2}(x)) = 0$. Note that $\mathbb{B}_{\delta/2}(k) \subseteq \mathbb{B}_{\rho(x, k) + \delta/2}(x)$. Recall that the support of μ is K . By definition, it follows $\mu(\mathbb{B}_{\delta/2}(k)) > 0$. Contradicting the monotonicity of the measure.
2. $S \geq \rho(x; K)$: By definition of supremum, given any $\epsilon > 0$, $\mu(\mathbb{B}_{S+\epsilon}(x)) > 0$. In other words, there exists some x_ϵ such that $\rho(x, x_\epsilon) \leq S$ and $\mathbb{B}_\epsilon(x_\epsilon) > 0$. Again, μ being supported on K means that $x_\epsilon \in K$ so one sees that $\rho(x; K) = \inf_{y \in K} \rho(x, y) \leq \rho(x, x_\epsilon) \leq S$.

Thus finishing the proof. ♡

Algorithm 2 Estimate distance function through Invariant Measure method

Input: $x_0 \in S$; $n \in \mathbb{N}$ (Size of orbit of x_0); $\{f_1, \dots, f_m\}$ a contractive IFS; $x \in S$ a point; $\epsilon > 0$ an error tolerance.

Output: An estimate of $\rho(x; K)$

Compute the orbit $\{x_0, x_1, \dots, x_n\}$ as defined in Theorem 4.17

Let $R_{\min} = 0$, $R_{\max} = E$ where E is a large enough such that $K \subseteq \mathbb{B}_E(x)$

while $R_{\max} - R_{\min} > \epsilon$ **do**

Estimate $\mu(\mathbb{B}_{R_{\max}}(x))$

if $\mu(\mathbb{B}_{R_{\max}}(x)) > 0$ **then**

$R_{\max} \leftarrow R_{\min} + (R_{\max} - R_{\min})/2$

else

$R_{\min} \leftarrow R_{\min} + (R_{\max} - R_{\min})/2$

end if

end while

return $(R_{\max} - R_{\min})/2$

Despite how nice this theory may be, this algorithm is of little practical value, because for a high resolution to be achieved, the size of the orbit needed to estimate the measure becomes too memory-demanding. As such we don't consider it any further.

We now present another method to Ray-March the Attractor of an IFS. Algorithmically, this will resemble the Region Subdivision algorithm, but in contrast to all the methods presented until now, we will not try to give an accurate estimate of the distance to the attractor. Rather, we will exploit the nature of the IFS and the Ray-Marching algorithm itself to show that one can avoid the computation of the distance function whilst still producing *the same* image, provided that some special bounds for the distance can be found.

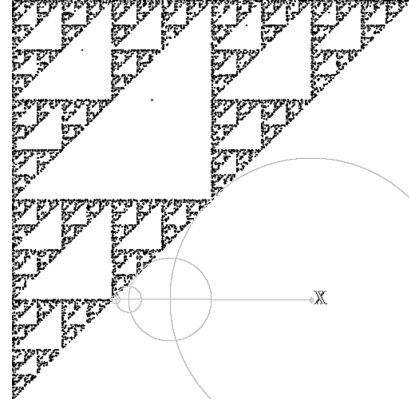


Figure 18: Simulation of Ray-Marching by estimating the distance function through the Invariant Measure method. The invariant measure of the Sierpinski Triangle is plotted according to Barnsley's Method. The radius of each ball is computed according to the presented algorithm.

5 Equivalent Distance Functions and Distance Underestimators

From now on we assume S is a complete, normed vector space while still using the notation $\rho(x, y)$ to denote the distance between two points x and y , which should be thought of as $\|x - y\|$. Let us introduce the following two definitions, which will be the main focus of our study in this section.

Definition 5.1 Let $f : S \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ be two functions. If there exists constants $0 < c \leq C$ such that for any $x \in S$:

$$cf(x) \leq g(x) \leq Cf(x)$$

We say $f(x)$ is **equivalent** to $g(x)$ and we write $f \sim g$. We call c, C the similarity constants.

Definition 5.2 Let $\Sigma \subseteq S$ be a non-empty subset. A **distance underestimator** to Σ is a non-negative function $f : S \rightarrow \mathbb{R}$ such that $f(x) \leq \rho(x; \Sigma)$.

Definition 5.2 was proposed by the authors of [Cou+22, Section 2.2] to develop their discussion of Ray-Marching. Throughout this section we will try to understand ways in which we can find distance underestimators to a set that are also equivalent to its distance function.

Definition 5.3 Let $\Sigma \subseteq S$ be a non-empty set and let $f : S \rightarrow \mathbb{R}$ be a distance underestimator to Σ that is also equivalent to its distance function $\rho(x; \Sigma)$. We say $f(x)$ is an **equivalent distance function** to Σ .

Our motivation studying these objects is very similar to that of the authors of [Cou+22]: namely that once we establish that a function f is an equivalent distance function to some set $\Sigma \subseteq S$, we will replace $\rho(x; \Sigma)$ with $f(x)$ in the Ray-Marching algorithm and after some minor adjustments, hopefully still be able to Ray-March Σ *without losing too much information*. As we will show, if $f(x)$ is equivalent to $\rho(x; \Sigma)$, in fact no information is lost at all.

Remark 5.4 The added power of a function f being equivalent to the distance function is that $f(x)$ becomes arbitrarily small if and only if x becomes arbitrarily close to Σ . This will turn out to be key to perform Ray-Marching without losing any information about Σ , since a function that is purely a distance underestimator can produce *false collisions*, see Figure 23. This weakness of distance underestimators was noted by the authors of [Cou+22], who suggested that having a second bound (i.e: obtaining an equivalent distance function) would prove beneficial in practice. On the other hand, obtaining a distance underestimator is far easier than obtaining an equivalent distance function, and for practical purposes, distance underestimators are good enough.

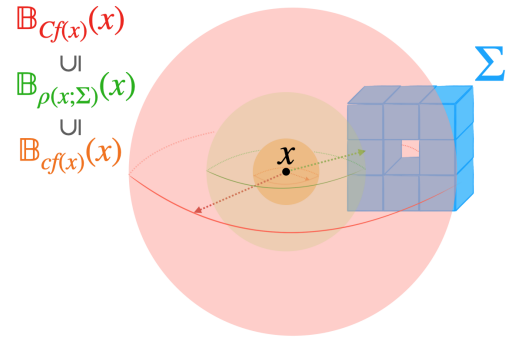


Figure 19: An illustration of the concept of equivalent distance functions in terms of open balls

Proposition 5.5 The relationship $\sim$ as defined in Definition 5.1 is an equivalence relation.

Proof. Routine checking:

1. Let $f \sim g$ with similarity constants c, C : Note that $cf \leq g \leq Cf = \frac{C}{c}cf \leq \frac{C}{c}g$. Hence $\frac{1}{C}g \leq f \leq \frac{1}{c}g$, so $g \sim f$ with similarity constants C^{-1}, c^{-1} .
2. Clearly $f \sim f$ with similarity constants $c = C = 1$.
3. Let $f \sim g$ with similarity constants c_1, C_1 and $g \sim h$ with similarity constants c_2, C_2 . Then $c_1c_2f \leq c_2g \leq h \leq C_2g \leq C_1C_2f$. Thus $f \sim h$ with similarity constants c_1c_2, C_1C_2 .

♡

5.1 The Ray-Marching function and Essentially Equal Ray-Marching

When discussing Ray-Marching, it was said that the algorithm produces a picture of a set Σ on a computer screen. To give a formal treatment of the algorithm, we can turn it into a more mathematical idea, and treat the Ray-Marching algorithm as a function, whose image is a set containing Σ . Namely, the set of points that would be labeled by the Ray-Marching algorithm to be in Σ up to ϵ collision tolerance.

Definition 5.6 Let f be a non-negative function. The Ray-Marching function is a map $\mathcal{R}_f : (0, \infty) \rightarrow \mathcal{P}(S)$ defined by:

$$\mathcal{R}_f : \epsilon \mapsto \{x \in S \mid f(x) < \epsilon\}$$

Where we take $\mathcal{P}(S)$ to be the set of subsets of S .

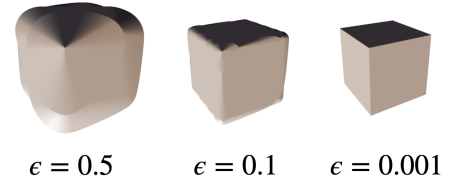


Figure 20: Ray-Marching of a cube.

An important observation is that the Ray-Marching function is monotonic in ϵ , in the sense that if $\epsilon_1 < \epsilon_2$, then $\mathcal{R}_f(\epsilon_1) \subseteq \mathcal{R}_f(\epsilon_2)$. This observation can be put into the following proposition, which will serve as a motivation for the continuation of our discussion:

Proposition 5.7 Let (S, ρ) be $(\mathbb{R}^d, d_e)$ and let $\Sigma \in \mathbb{H}(S)$ with $f : S \rightarrow \mathbb{R}$ an equivalent distance function to $\rho(x; \Sigma)$. Let (ϵ_n) be a monotonically decreasing sequence converging to zero. Then the sequence of sets

$$\overline{\mathcal{R}_f(\epsilon_0)} \supset \overline{\mathcal{R}_f(\epsilon_1)} \supseteq \dots$$

converges in the Hausdorff metric to $\bigcap_{n \geq 0} \overline{\mathcal{R}_f(\epsilon_n)}$. Where $\overline{A}$ denotes the closure of A .

Proof. First we note that each since $cf(x) \leq \rho(x; \Sigma) \leq Cf(x)$, one has that $\mathcal{R}_f(\epsilon) \subseteq \mathcal{R}_{\rho(x; \Sigma)}(C\epsilon)$ which is precisely equal to $N_{C\epsilon}(\Sigma)$. Since this set is bounded, it follows that each $\overline{\mathcal{R}_f(\epsilon_n)}$ is closed and bounded hence compact. Now all left to show is that a decreasing sequence A_n of compact sets converges to $A := \bigcap_n A_n$ with respect to the Hausdorff metric. We adapt the proof given in Kigami's

Book [Kig01, Page 11]. Since A is a subset of each A_n , it follows that given any $\epsilon > 0$, $A \subseteq N_\epsilon(A_n)$. We now need to show that for large enough n , $A_n \subseteq N_\epsilon(A)$. Suppose that this wasn't the case, then for all n , and any $x \in A_n$, we have $\rho(x; A) \geq \epsilon$. Form a sequence (x_n) by taking each x_n to be any point in A_n . Since A_1 is compact and contains every other A_n , there exists a converging subsequence (x_{n_k}) the limit of this sequence, x' belongs to A , thus contradicting our assumption, since for large enough n , the elements of the sequence (x_{n_k}) will become arbitrarily close to a point of A . $\heartsuit$

One can think that each term of the decreasing sequence of Proposition 5.7 is *removing noise*, and as such, the limit is in some sense the *idealised output* of the Ray-Marching algorithm, motivating Definition 5.8.

Definition 5.8 Let $f(x), g(x)$ be two non-negative functions. We say that they produce essentially equal Ray-Marching if:

$$\bigcap_{\epsilon > 0} \mathcal{R}_f(\epsilon) = \bigcap_{\epsilon > 0} \mathcal{R}_g(\epsilon)$$

The wait is over, we can finally reveal the result we have been hinting at throughout this section:

Theorem 5.9 Equivalent non-negative functions $f(x)$ and $g(x)$ produce essentially equal Ray-Marching.

Proof. We show $\bigcap_{\epsilon > 0} \mathcal{R}_f(\epsilon) \subseteq \bigcap_{\epsilon > 0} \mathcal{R}_g(\epsilon)$.

Let any $\epsilon > 0$ be given and $x \in \bigcap_{\epsilon' > 0} \mathcal{R}_f(\epsilon')$. We aim to show $x \in \mathcal{R}_g(\epsilon)$. Recall that from equivalence $cf(x) \leq g(x) \leq Cf(x)$, so in particular $\frac{1}{C}g(x) \leq f(x)$. Thus if we need $g(x) < \epsilon$, it is sufficient to ask that $f(x) < \frac{1}{C}\epsilon$, as then one would have $\frac{1}{C}g(x) < \frac{1}{C}\epsilon$. Thus we just set $\epsilon' = \frac{1}{C}\epsilon$ and it follows $x \in \mathcal{R}_g(\epsilon)$. The choice of ϵ was arbitrary so it follows $x \in \bigcap_{\epsilon > 0} \mathcal{R}_g(\epsilon)$ thus establishing the first inclusion.

The second inclusion follows by a symmetric argument. $\heartsuit$

A very similar result to Theorem 5.9 was shown in [Cou+22, Lemma 2.5], where the authors show that distance underestimators also produce the same image as the true distance function when taking the error tolerance $\epsilon \rightarrow 0$. However, here we see the advantage of using equivalent distance functions instead of underestimators: one can explicitly instruct how to change their error and escape tolerances ϵ and E , to produce the same image as would be produced by Ray-Marching via the distance function:

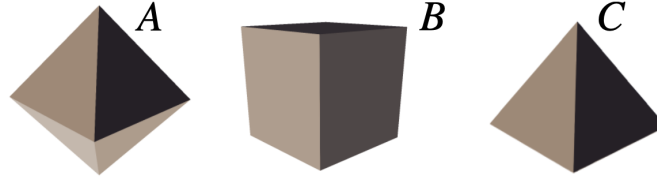
Remark 5.10 By following the proof of Theorem 5.9, one can modify the original Ray-Marching algorithm to work with equivalent distance functions. Indeed, if we have $cf(x) \leq \rho(x; \Sigma) \leq Cf(x)$, we will modify the step:

$$\text{Move } x \rightarrow x' = x + \rho(x; \Sigma)d \quad \text{To:} \quad \text{Move } x \rightarrow x' = x + cf(x)d$$

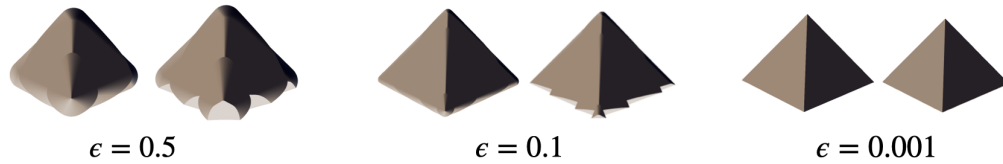
Where d was the specified direction to Ray-March in. Moreover, instead of checking whether

$$\text{while } \epsilon < \rho(x; \Sigma) < E \quad \text{we check:} \quad \text{while } \frac{\epsilon}{c} < f(x) < \frac{\epsilon}{C} :$$

Example 5.11 Let us illustrate Theorem 5.9 with a concrete example: Consider the sets A, B, C as shown in the diagram below, where C is thought of as the intersection $A \cap B$. We can explicitly compute each set's distance function (see [Quia]), but alternatively, we can come up with an equivalent distance function to C by combining $\rho(x; A)$ and $\rho(x; B)$.



Indeed, as we will see in Theorem 5.18, it turns out that $g(x) := \rho(x; C) = \rho(x; A \cap B)$ and $f(x) := \max\{\rho(x; A), \rho(x; B)\}$ are equivalent distance functions (and of course also $f(x)$ underestimates $g(x)$). Consider the sets $\mathcal{R}_f(\epsilon)$ and $\mathcal{R}_g(\epsilon)$ viewed from the same angle with different values for the ϵ tolerance.



One can see that for large values of ϵ , the sets generated are different, but as one takes ϵ to be very small, they become indistinguishable. The huge advantage of using distance underestimators or equivalent distance functions is that even if we didn't know $\rho(x; C)$, we could have still produce an *essentially equal* image provided we knew $\rho(x; A)$ and $\rho(x; B)$. Notice that if $g(x)$ was simply an underestimator, this example would also work perfectly well, as we are effectively just taking error tolerances to zero.

We now move on to explore applications of equivalent distance functions and distance underestimators. In particular, we will see how given a collection of sets and their distance functions, one can construct a wide variety of distance underestimators and equivalent distance functions which we can use to Ray-March sets that may have not been able to be Ray-Marched using only the previously available information. After our exploration, we will combine the results obtained and return to our initial goal of Ray-Marching the attractor of an IFS. For the rest of the discussion on equivalence of distance functions and distance underestimators, we will focus on trying to obtain the stronger results given by equivalence, and only when not possible, resort to underestimating.

5.2 Equivalence under invertible transformations

The first question we want to ask is the following: suppose one is given a non-empty set $\Sigma \subset S$ with its distance function, as well as a transformation $f : S \rightarrow S$. Can we Ray-March the set $f[\Sigma]$? Unfortunately, this question is too general, but imposing some additional structure on our transformation will allow us to solve the problem.

Definition 5.12 Let $f : S \rightarrow S$ be a transformation. We say f is **regular** if there are constants $0 < a < b$ such that:

$$a\rho(x, y) \leq \rho(f(x), f(y)) \leq b\rho(x, y)$$

Informally, a regular transformation is just a transformation that does not excessively contract or expand space. These upper and lower bounds for our “distortion” of space is what will give us equivalence.

Theorem 5.13 Let $f : S \rightarrow S$ be a regular invertible transformation and $\Sigma \subseteq S$ be a non-empty set. Then the distance function $\rho(x; f[\Sigma])$ and $\rho(f^{-1}(x); \Sigma)$ are equivalent.

Proof. Writing $\rho(x, f(y)) = \rho(f \circ f^{-1}(x), f(y))$ and using the fact that f is regular, we deduce that:

$$a\rho(f^{-1}(x), y) \leq \rho(x, f(y)) \leq b\rho(f^{-1}(x), y)$$

Where a and b are as in Definition 5.12. Now, since taking the infimum of y over Σ preserves the inequalities, one sees:

$$a\rho(f^{-1}(x); \Sigma) \leq \rho(x; f[\Sigma]) \leq b\rho(f^{-1}(x); \Sigma)$$

As required. ♡

This technique of considering the distance from $f^{-1}(x)$ to Σ instead of the distance from x to $f[\Sigma]$ is already well known, at least for the case of similarities, see [Quia, Positioning]. In this article, Inigo notes that if f is a non-uniform scaling, the distance must be bounded and cannot be expressed exactly and as such gives no further consideration to these transformations. However, as shown by this theorem, this principle extends perfectly fine for any choice of f that behaves *well enough*.

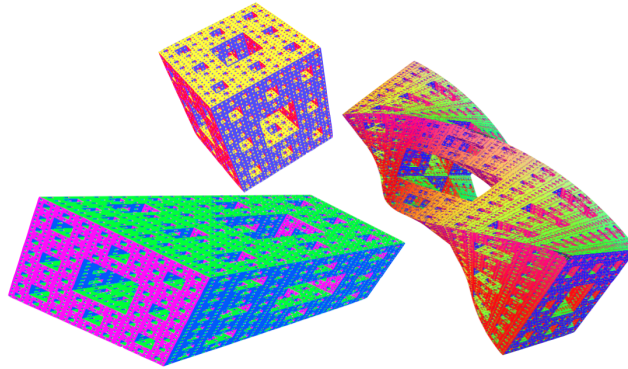


Figure 21: Ray-Marching of $\mathcal{M}$, $f[\mathcal{M}]$ (bottom left), and $g[\mathcal{M}]$ (right), where f is a non-uniform scaling and g is a *twisting* transformation.

5.3 Equivalence under Boolean set operations

Next in our exploration of applications of equivalent distance functions, we wish to tackle the next problem that arises naturally when attempting to Ray-March an IFS attractor. Recall that we thought of an attractor K as being a set that satisfied an equation of the sort:

$$K = \bigcup_{i=1}^m f_i[K].$$

We have already discussed how to tackle the Ray-Marching of $f[\Sigma]$ for a special kind of transformations f , but if we wish to Ray-March K , we need to understand how to Ray-March unions of sets. In this section, we tackle the more general problem of understanding the Ray-Marching of Boolean set operations, i.e: suppose A and B are non-empty subsets of S which we know how to Ray-March, how can one understand the Ray-Marching of: $A \cup B$, $A \cap B$ and $A \setminus B$?

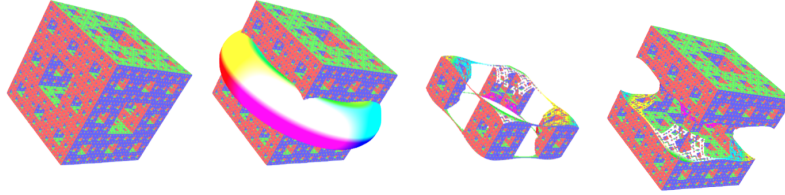


Figure 22: Ray-Marching of $\mathcal{M}$, $\mathcal{M} \cup T$, $\mathcal{M} \cap T$ and $\mathcal{M} \setminus T$, where T is a torus.

5.3.1 Union of sets

Theorem 5.14 Let A, B be non-empty subsets of S . Then $\rho(x; A \cup B)$ and $\min\{\rho(x; A), \rho(x; B)\}$ are equivalent distance functions.

Actually, something even better is true, we have that the similarity constants are $c = C = 1$, in other words, *they are the same function!*

Proof. Let $x \in S$ be fixed. If it were the case that $\rho(x; A \cup B) < \min\{\rho(x; A), \rho(x; B)\}$ then there would exist a $y \in A \cup B$ with $\rho(x; A \cup B) < \rho(x, y) < \min\{\rho(x; A), \rho(x; B)\}$. The last inequality implies that $\rho(x, y) < \inf_{z \in A} \rho(x, z)$ and $\rho(x, y) < \inf_{w \in B} \rho(x, w)$. By assumption y belongs to A or B . Either of these cases contradicts the definition of infimum. It follows $\rho(x; A \cup B) \geq \min\{\rho(x; A), \rho(x; B)\}$

Conversely, suppose that $\rho(x; A \cup B) > \min\{\rho(x; A), \rho(x; B)\}$ then either $\rho(x; A \cup B) > \rho(x; A)$ or $\rho(x; A \cup B) > \rho(x; B)$. As mentioned in a previous proof: if $R \subset S$ then $\rho(x; R) \geq \rho(x; S)$. So either of the two stipulated cases leads to a contradiction, for $A \subset A \cup B$ and $B \subset A \cup B$. This shows $\rho(x; A \cup B) \leq \min\{\rho(x; A), \rho(x; B)\}$.

So in fact $\rho(x; A \cup B) = \min\{\rho(x; A), \rho(x; B)\}$. So one has that $\rho(x; A \cup B) \sim \min\{\rho(x; A), \rho(x; B)\}$ with both similarity constants being equal to 1. ♡

5.3.2 Intersection of sets

We have just seen how taking the union of two sets corresponds to taking the minimum of two distance functions. It would be really nice if the analogy extended so that taking the intersection of two sets corresponded to taking the maximum of two distance functions. Generally, this is incorrect, but some partial results can be obtained.

Lemma 5.15 Let A and B be non-empty subsets of S , then $\max\{\rho(x; A), \rho(x; B)\}$ is a distance underestimator of $\rho(x; A \cap B)$.

Proof. Clearly, if $S \subset R$ where S, R are non-empty subsets of S , one has that for $x \in S \setminus R$, $\rho(x; S) \geq \rho(x; R)$. In particular, this means that $\rho(x; A \cap B) \geq \rho(x; A)$ and $\rho(x; A \cap B) \geq \rho(x; B)$. Joining these two statements, it must follow: $\rho(x; A \cap B) \geq \max\{\rho(x; A), \rho(x; B)\}$. $\heartsuit$

Remark 5.16 As discussed previously, this result would in practice be sufficient to generate images that *look* as we would expect them to, but we would like to see if we can do a bit better. In particular, we would like to see under which conditions one can say that the functions in Lemma 5.15 are actually equivalent.

We will now add extra structure to our space S , in particular, we will take S to be a complete inner product space.

Definition 5.17 Given elements $a, b \in S$ we define:

- $[a, b] := \{y \in S | y = \mu a + (1 - \mu)b \quad 0 \leq \mu \leq 1\}$, the line between the points a and b .
- $r_b(a) := \{y \in S | y = a + \mu b \quad \mu \geq 0\}$, the ray emanating from point a in the direction of vector b .

A subset Σ of S is said to be **convex**, if given any two points $x, y \in \Sigma$, the line $[x, y]$ remains in S . I.e: If $z \in [x, y]$ implies $z \in \Sigma$.

Theorem 5.18 Let A, B be non-empty, convex and bounded subsets of S . Then the functions $\max\{\rho(x; A), \rho(x; B)\}, \rho(x; A \cap B)$ are equivalent.

Remark 5.19 (On the condition of convexity) . Let us illustrate why convexity is necessary. Consider the sets A and B as in the diagram and suppose we had equivalence:

$$c \max\{\rho(x; A), \rho(x; B)\} \leq \rho(x; A \cap B) \leq C \max\{\rho(x; A), \rho(x; B)\}.$$

We can choose the point x and the sets A and B so that $\max\{\rho(x; A), \rho(x; B)\} \leq \epsilon/C$. This means $\rho(x; A \cap B) \leq \epsilon$. However, without loss of generality we can choose A and B so that $\rho(x; A \cap B) > \epsilon$, contradicting the equivalence.

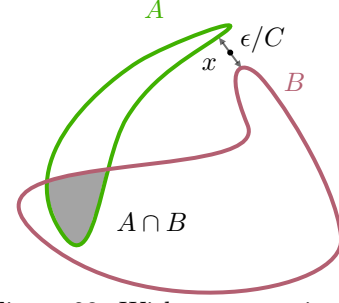


Figure 23: Without convexity one may encounter false collisions

The proof of Theorem 5.18 boils down to showing there is some constant C such that $\rho(x; A \cap B)$ is less than or equal to $C \max\{\rho(x; A), \rho(x; B)\}$. This result was shown by Hoffman in [Hof92, Lemma 2], who showed the result held without requiring both sets to be bounded. The proof we present is an adaptation of Hoffman's arguments to fit the context of our discussion, we will focus on sketching the main ideas of his argument and in particular, why convexity helps solve our problem. For a fully detailed explanation, the reader is invited to consult Hoffman's original proof.

Lemma 5.20 Let $A, B \subseteq S$ be non-empty convex bounded sets and let:

$$0 < \lambda_A < \sup_{y \in A \cap B} \rho(y, S \setminus A) \quad 0 < \lambda_B < \sup_{y \in A \cap B} \rho(y, S \setminus B)$$

Then for any $x \in S$, the distance $\rho(x; A \cap B)$ is less than or equal to:

$$\max \left\{ \frac{\text{diam}(A) - \lambda_A}{\lambda_A}, \frac{\text{diam}(B) - \lambda_B}{\lambda_B} \right\} \max\{\rho(x; A), \rho(x; B)\}.$$

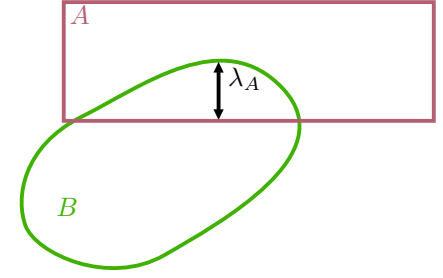


Figure 24: λ_A in Lemma 5.20

We refer to this condition on λ_A and λ_B as the escape criterion, and it should be thought of a lower bound of the maximum distance one can be forced to travel starting from $A \cap B$ in order to escape A or B respectively. Observe that Lemma 5.20 will automatically prove Theorem 5.18, but first we need a simpler result:

Lemma 5.21 Let A, B, λ_A be as in Lemma 5.20 and let $x \in A$. Then $\rho(x; A \cap B) \leq \frac{\text{diam}(A) - \lambda_A}{\lambda_A} \rho(x; B)$.

Sketch of proof of Lemma 5.21. If $\rho(x; B) = \rho(x; A \cap B)$ then the claim holds by noting that $\lambda_A \leq \text{diam}(A)/2$. Assume $\rho(x; B) < \rho(x; A \cap B)$. Notice that by construction of λ_A , we can find a $z \in A \cap B$ such that $\mathbb{B}_{\lambda_A}(z) \subset A$. Moreover, for any $\epsilon > 0$ given, we can find an $x_0(\epsilon) \in B$ such that $\rho(x; B) \leq \rho(x, x_0) \leq \rho(x; B) + \epsilon$.

The proof continues by constructing a plane Π spanned by the vectors $x - x_0(\epsilon)$ and $x - z$ that passes through the point x . However, it is not clear from our construction whether these vectors should be linearly independent. In the proof of [Hof92, Lemma 2], the author gets around this by considering a supporting hyperspace H of B . After showing the claim holds for the sets A and H , one can relate this back to the original set B . Since we are interested in seeing how convexity is used, rather than on the topological intricacies of this proof, we will proceed by assuming that the vectors $x - x_0(\epsilon)$ and $x - z$ are linearly independent.

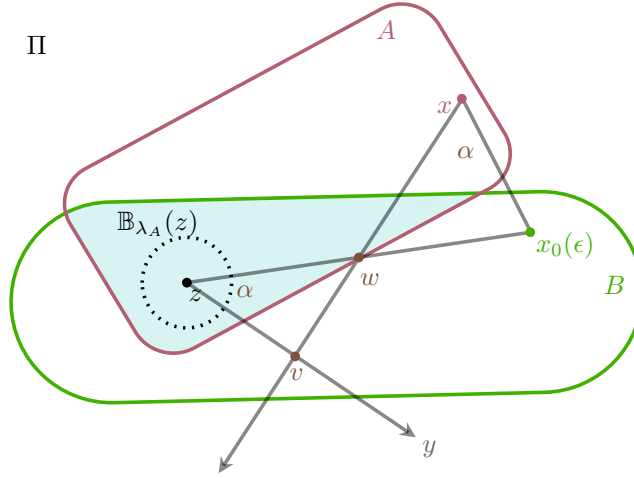


Figure 25: Illustration of the sketch of the proof of Lemma 5.21.

Let Π be the plane spanned by the points $x, z, x_0(\epsilon)$. We now proceed to construct similar triangles in this plane. To do so consider the point $w = [z, x_0(\epsilon)] \cap \partial \bar{A} \cap \Pi$, and through the use of the inner product, determine the angle α held by the vectors $w - x$ and $x_0(\epsilon) - x$. Let $y \in \Pi$ be a vector such that the angle between $w - z$ and y is also α . By fixing $v = (r_{w-x}(x) \setminus [x, w]) \cap r_y(z)$, we have now constructed similar triangles $\Delta(x, w, x_0(\epsilon))$ and $\Delta(w, z, v)$. Observe that:

1. $\rho(x, w) \geq \rho(x; A \cap B)$ (This is because $w \in [z, x_0(\epsilon)] \cap A$, so by convexity $w \in B$ and $w \in A$).
2. $\rho(x, x_0(\epsilon)) \leq \rho(x; B) + \epsilon$ (By construction).
3. $\rho(z, w) \leq \text{diam}(A) - \lambda_A$ (Since $\rho(z, w) < \sup_{y \in A \cap B} \rho(y; S \setminus A) \leq \text{diam}(A)/2$).
4. $\rho(z, w) \geq \lambda_A$ (This is because $\mathbb{B}_{\lambda_A}(z)$ is contained in the interior of A and by convexity, v is not).

By similarity of triangles, we know that:

$$\frac{\rho(x, x_0)}{\rho(x, w)} = \frac{\rho(z, v)}{\rho(z, w)}$$

Using the facts above, we reach the conclusion $\rho(x; A \cap B) \leq \frac{\text{diam}(A) - \lambda_A}{\lambda_A} (\rho(x; B) + \epsilon)$. Taking $\epsilon \rightarrow 0$ the Lemma is proven.

♡

Proof of Lemma 5.20. We distinguish different cases:

- $x \in A \setminus B$: By Lemma 5.21, we have $\rho(x; A \cap B) \leq \frac{\text{diam}(A) - \lambda_A}{\lambda_A} \rho(x; B)$ which is clearly less than $\max \left\{ \frac{\text{diam}(A) - \lambda_A}{\lambda_A}, \frac{\text{diam}(B) - \lambda_B}{\lambda_B} \right\} \max\{\rho(x; A), \rho(x; B)\}$.
- $x \in B \setminus A$. Given that we assume both A and B to be bounded Lemma 5.21 also applies and by the same argument as the previous point, the theorem holds.
- $x \in A \cap B$, then $\rho(x; A \cap B) = \rho(x; A) = \rho(x; B) = 0$ so the theorem holds trivially.
- $x \notin A \cup B$. If $\rho(x; A \cap B) = \rho(x; B)$, then due to the bounds λ_A, λ_B are less than $\text{diam}(A)/2, \text{diam}(B)/2$ respectively, the theorem holds trivially. Hence assume $\rho(x; A \cap B) > \rho(x; B)$. The proof proceeds as follows:

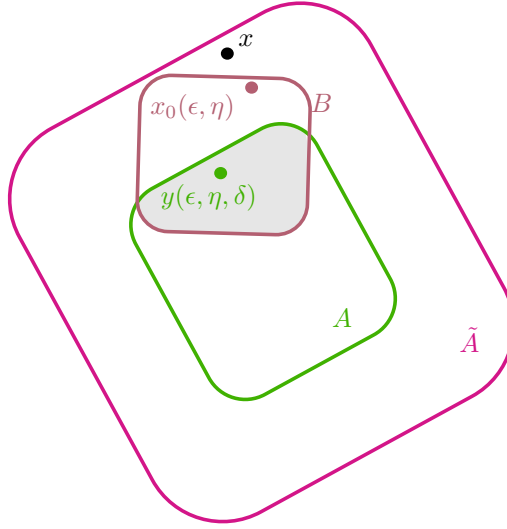


Figure 26: Illustration of the fourth part of the proof of Lemma 5.20

Let $\eta > 0$ and let $\tilde{A} := N_{\rho(x; A) + \eta}(A)$ be the $\rho(x; A) + \eta$ Hausdorff Neighborhood of A . This ensures now that $x \in \tilde{A}$. Our goal is to be able to apply Lemma 5.21, so we need to find some $\lambda_{\tilde{A}}$ that satisfies the escape criterion for $\tilde{A}$. Informally speaking, the escape criterion is an upper bound for the maximum distance needed to escape $\tilde{A}$ starting from $B \cap \tilde{A}$. Using this intuition it is clear to see that

$$\lambda_{\tilde{A}} := \sup_{y \in A \cap B} \rho(y; S \setminus A) + \rho(x; A) + \eta < \sup_{y \in \tilde{A} \cap B} \rho(y; S \setminus \tilde{A}).$$

So we see this choice of $\lambda_{\tilde{A}}$ satisfies the escape criterion so we are in good shape to apply Lemma 5.21. Furthermore, note that $\text{diam}(\tilde{A}) = \text{diam}(A) + 2\rho(x; A) + 2\eta$. Let now $\epsilon > 0$ be given, by definition of $\rho(x; \tilde{A} \cap B)$, there exists some $x_0(\epsilon, \eta) \in \tilde{A} \cap B$ such that:

$$\rho(x, x_0(\epsilon, \eta)) \leq \rho(x; \tilde{A} \cap B) + \epsilon \leq \frac{\text{diam}(\tilde{A}) - \lambda_{\tilde{A}}}{\lambda_{\tilde{A}}} \rho(x; B) + \epsilon.$$

By assumption of $\rho(x; A \cap B) > \rho(x; B)$, we can also choose $x_0(\epsilon)$ such that it does not belong to $A \cap B$. Given that B is bounded, we can use Lemma 5.21 to deduce $\rho(x_0(\epsilon, \eta); A \cap B) \leq \frac{\text{diam}(B) - \lambda_B}{\lambda_B} \rho(x_0(\epsilon, \eta); A)$

which is trivially less than $\frac{\text{diam}(B)-\lambda_B}{\lambda_B}(\rho(x; A) + \eta)$. Now by definition of $\rho(x_0(\epsilon, \eta); A \cap B)$, we see that given any $\delta > 0$, one can find some $y(\epsilon, \eta, \delta) \in A \cap B$ with:

$$\rho(x_0(\epsilon, \eta), y(\epsilon, \eta, \delta)) \leq \rho(x_0(\epsilon); A \cap B) + \delta.$$

Putting it all together:

$$\rho(x; A \cap B) \leq \rho(x, y(\epsilon, \eta, \delta)) \leq \rho(x, x_0(\epsilon, \eta)) + \rho(x_0(\epsilon, \eta), y(\epsilon, \eta, \delta)).$$

Hence it follows:

$$\rho(x; A \cap B) \leq \frac{\text{diam}(\tilde{A})-\lambda_{\tilde{A}}}{\lambda_{\tilde{A}}} \rho(x; B) + \frac{\text{diam}(B)-\lambda_B}{\lambda_B} [\rho(x; A) + \eta] + \epsilon + \delta$$

Simplifying this expression with the values of $\lambda_{\tilde{A}}$ and $\text{diam}(\tilde{A})$ we obtain that

$$\rho(x; A \cap B) \leq \frac{\text{diam}(A)+\rho(x; A)-\lambda_A+\eta}{\lambda_A+\rho(x; A)} \rho(x; B) + \frac{\text{diam}(B)-\lambda_B}{\lambda_B} [\rho(x; A) + \eta] + \epsilon + \delta.$$

Combining this with the fact that for real numbers $a > b > 0$, if $c \geq 0$ then $\frac{a+c}{b+c} \leq \frac{a}{b}$ and taking the limits $\epsilon, \eta, \delta \rightarrow 0$ we obtain:

$$\rho(x; A \cap B) \leq \frac{\text{diam}(A)-\lambda_A}{\lambda_A} \rho(x; B) + \frac{\text{diam}(B)-\lambda_B}{\lambda_B} \rho(x; A).$$

It is now easy to see that $\rho(x; A \cap B) \leq \max \left\{ \frac{\text{diam}(A)-\lambda_A}{\lambda_A}, \frac{\text{diam}(B)-\lambda_B}{\lambda_B} \right\} \max\{\rho(x; A), \rho(x; B)\}$. ♡

Remark 5.22 In [Mar04], Martínón showed that if you impose even stricter conditions on A, B , namely $A \cup B$ being convex, then $\rho(x; A \cap B)$ is actually equal to $\max\{\rho(x; A), \rho(x; B)\}$. Sadly this condition is too strict and doesn't even hold in any of the sets constructed in this project.

5.3.3 Set complements

Suppose A and B are two non-empty sets, how can we estimate $\rho(x; A \setminus B)$? The answer involves the introduction of a **Signed Distance Function**.

Definition 5.23 Let $A \subset S$ be a non-empty set and $x \in S$ be a point. We define the Signed Distance Function of A

$$\sigma_A(x) := \begin{cases} \rho(x; A) & x \in S \setminus A \\ -\rho(x; S \setminus A) & x \in A \end{cases}$$

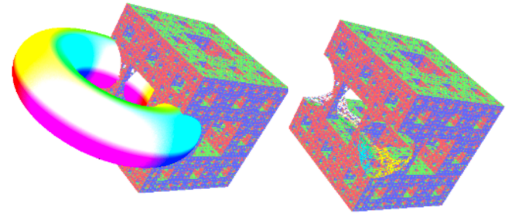


Figure 27: Ray-Marching of Menger's Sponge union and complement a torus

In plain English, the signed distance function is a positive distance if the point is outside the set and negative if it is in the set. This gives information about the inside of the set as opposed to regular distance functions, which kill all the information once you step inside the set. This allows us to concoct a distance underestimator.

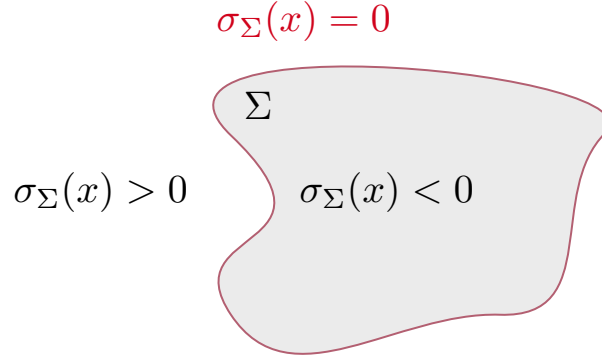


Figure 28: Illustration of the signed distance function

Theorem 5.24 Let A, B be non-empty sets. Then $\max\{\rho(x; A), -\sigma_B(x)\}$ is a distance underestimator of $\rho(x; A \setminus B)$

Proof. First of all, notice that $\rho(x; A \setminus B) = \rho(x; A \cap S \setminus B)$. Now, note that we have the distance underestimator $\max\{\rho(x; A), \rho(x; S \setminus B)\} \leq \rho(x; A \cap B)$. Now observe that from the definition of the Signed Distance Function, $-\sigma_B(x) = \sigma_{S \setminus B}(x)$. Therefore, if $x \in S \setminus B$, then $\max\{\rho(x; A), -\sigma_B(x)\} = \max\{\rho(x; A), \rho(x; S \setminus B)\} \leq \rho(x; A \setminus B)$. On the other hand, if $x \in B$, then $\max\{\rho(x; A), -\sigma_B(x)\} = \max\{\rho(x; A), -\rho(x; B)\} = \rho(x; A) \leq \rho(x; A \setminus B)$. $\heartsuit$

Remark 5.25 Could we possibly hope for equivalence of distance functions in this scenario? Unfortunately this is not the case. The reason is that in the above discussion we have effectively shown that $\max\{\rho(x; A), \rho(x; S \setminus B)\}$ is a distance underestimator for $\rho(x; A \setminus B)$, and if we wanted equivalence, we would need $S \setminus B$ to be convex, which generally will not be the case.

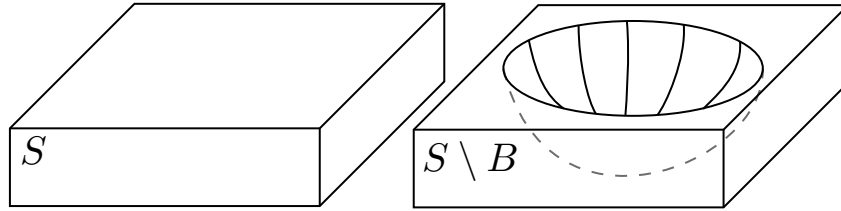


Figure 29: Illustration of Remark 5.25

5.4 Equivalence under limits

In sections 5.2 and 5.3 we have discussed how to Ray-March regular invertible transformations of sets, as well as their unions, intersections, and complements. The missing ingredient is the following: recall that the attractor K of an IFS is defined in a self-referential way, i.e: $K = \bigcup_i f_i[K]$, so even if we apply the techniques we have just developed to attempt to find an equivalent distance function to K , we would end up with some function defined in terms of $\rho(x; K)$. To bypass this issue, we can use the fact that from Remark 3.18, we know that K can also be obtained as the limiting operation of repeatedly applying the Hutchinson operator T to any initial set $K_0 \in \mathbb{H}(S)$. In other words, all left to know is how to deal with equivalence of distance functions under limiting operations.

Lemma 5.26 Let (f_n) and (g_n) be two sequence of functions such that for any $n \in \mathbb{N}$, one has $f_n \sim g_n$ with $c_n g_n(x) \leq f_n(x) \leq C_n g_n(x)$, such that $0 < \inf\{c_n \mid n \geq 1\} \leq \sup\{C_n \mid n \geq 1\} < \infty$ Then the functions:

$$F(x) := \liminf_{n \rightarrow \infty} f_n(x) \quad G(x) := \liminf_{n \rightarrow \infty} g_n(x)$$

are equivalent.

Proof. Since for any $n \in \mathbb{N}$, and any x in the domain, we have that $c_n g_n(x) \leq f_n(x) \leq C_n g_n(x)$, it is clear that in particular, the following inequality holds:

$$\inf\{c_n \mid n \geq 1\} g_n(x) \leq f_n(x) \leq \sup\{C_n \mid n \geq 1\} g_n(x)$$

Taking the limit infimum preserves inequalities, so we see:

$$\inf\{c_n \mid n \geq 1\} G(x) \leq F(x) \leq \sup\{C_n \mid n \geq 1\} G(x)$$

Thus we reach the conclusion $F \sim G$. ♡

Corollary 5.27 Let $\{f_1, \dots, f_m\}$ and $\{g_1, \dots, g_m\}$ be two finite families of functions such that for each i , $f_i \sim g_i$. Then the functions $F(x) := \min_{1 \leq i \leq m} \{f_i(x)\}$ and $G(x) := \min_{1 \leq i \leq m} \{g_i(x)\}$ are equivalent.

Proof. Define the new sequences (f'_n) and (g'_n) where $f'_n = f_{(n \bmod m)}$ and $g'_n = g_{(n \bmod m)}$. Using the previous lemma, we see that $F(x)$ coincides with $\liminf_{n \rightarrow \infty} f'_n(x)$ and similarly for $G(x)$, so we deduce $F \sim G$. ♡

Remark 5.28 Suppose one has two sequences (f_n) and (g_n) of functions satisfying the conditions of Lemma 5.26. If we suppose additionally assume both sequences of functions have pointwise limits $f(x)$ and $g(x)$, it follows that these limits are also equivalent, i.e: $f \sim g$. One can think of this result in terms of the following “commutative diagram”.

$$\begin{array}{ccc} f_n & \xrightarrow{\text{pointwise}} & f \\ \downarrow \sim & & \downarrow \sim \\ g_n & \xrightarrow{\text{pointwise}} & g \end{array}$$

6 Equivalent Distance Function to The Attractor

Equipped with the technology of 5.2, 5.3 and 5.4, we are finally ready to Ray-March the attractor of an IFS through the method of equivalent distance functions. Unfortunately, the mission of visualising any attractor is going to be out of our reach, at least with the tools presented in section 5, and for a very similar reason as in 5.2, we will restrict our attention to IFS's which don't distort space excessively. For this, we introduce the following definition, which should hopefully look natural by now.

Definition 6.1 Let $f : S \rightarrow P$ be a function between metric spaces (S, ρ) and (P, σ) . We say f is a **regular contraction** if there exists constants $0 < a < b < 1$ such that:

$$a\rho(x, y) \leq \sigma(f(x), f(y)) \leq b\rho(x, y).$$

An IFS $\{f_1, \dots, f_m\}$ where each f_i is a regular contraction is called a **regular IFS**.

We now present the main result of this section:

Theorem 6.2 Let $\{f_1, \dots, f_m\}$ be a regular IFS. Let T be the corresponding Hutchinson operator and let K be the attractor of the IFS. The following statements hold:

1. There exists a sequence of functions $(d_n) : S \rightarrow \mathbb{R}$, whose pointwise limit $d : S \rightarrow \mathbb{R}$ is equivalent to the distance function of the attractor, i.e: $d \sim \rho(x; K)$.
2. Each $d_n(x)$ is equivalent to a function of $d_{n-1}(x)$, i.e: there is some function $\phi_n : S \rightarrow \mathbb{R}$ such that $\phi_n = \phi_n(x; d_{n-1})$ and $\phi_n \sim d_n$.

Remark 6.3 The “punchline” of this theorem, is that one can recursively build a sequence of functions which we will be able to explicitly compute, provided that we choose the initial term wisely. Finally, the pointwise limit of this sequence will be equivalent to the distance function of the attractor.

Proof. Let $x \in S$ be fixed and choose any $K_0 \in \mathbb{H}(S)$. By continuity of the metric, one has that as $n \rightarrow \infty$:

$$\rho(x; T^{\circ(n)}[K_0]) \rightarrow \rho(x; K).$$

The strategy is now to construct a converging sequence $(d_n) : S \rightarrow \mathbb{R}$, such that for each n , $d_n(x)$ is equivalent to $\rho(x; T^{\circ(n)}[K_0])$. By Remark 5.28 it will follow that the limit of this sequence, $d(x)$ will be equivalent to $\rho(x; K)$. We will construct this sequence recursively. The first term d_0 is defined to be $\rho(x; K_0)$. Now we construct the n^{th} term. For convenience let $K_n = T^{\circ(n)}[K_0]$:

$$\rho(x; K_n) = \rho(x; T[K_{n-1}]) = \rho\left(x; \bigcup_{i=1}^m f_i[K_{n-1}]\right) \sim \min_{1 \leq i \leq m} \{\rho(x; f_i[K_{n-1}])\}$$

Since each f_i is a regular transformation, Theorem 5.13 tells us that $\rho(x; f_i[K_n]) \sim \rho(f_i^{-1}(x), K_n)$. By Corollary 5.27, we conclude that $\rho(x; K_n) \sim \min_{1 \leq i \leq m} \{\rho(f_i^{-1}(x); K_{n-1})\}$. This last term is what we will

define to be our n^{th} term of the sequence for $n \geq 1$. Moreover note that since (f_i) is a finite family of regular transformations, one has $\inf\{a_i \mid i \geq 1\} > 0$ and $\sup\{b_i \mid i \geq 0\} < \infty$ so the conditions of Remark 5.28 apply. This means that the limit of (d_n) , if it exists, will be equivalent to $\rho(x; K)$. The limit does exist, indeed, since taking the minimum of a finite collection of continuous functions is in turn a continuous function, we conclude:

$$d_n(x) \rightarrow \min_{1 \leq i \leq m} \{\rho(f_i^{-1}(x); K)\}$$

Label this limit by $d(x)$. Part one of the theorem is now proven. The second part follows from the definition of $d_n(x)$. Indeed, observing that $\rho(f_i^{-1}(x); K_{n-1})$ is equivalent to $d_{n-1}(f_i^{-1}(x))$, one can apply once again Corollary 5.27 to see that $d_n \sim \min_{1 \leq i \leq m} d_{n-1}(f_i^{-1}(x)) =: \phi(x; d_{n-1})$, as required. $\heartsuit$

Remark 6.4 (Conditions of the theorem) Notice that in the proof of this theorem, the requirements of the IFS being regular were crucial. Indeed, if we only required the IFS to be a contractive IFS, the “trick” used in the proof of theorem 5.13, would only yield an overestimation of the distance function, which would not suffice to Ray-March the set.

The results of this theorem are summarised in the following “commutative diagram”:

$$\begin{array}{ccccc} \dots & \sim & \phi_n(x; d_{n-1}) & \sim & d_n(x) & \xrightarrow{\text{pointwise}} & d(x) \\ & & & & \vdots & & \vdots \\ & & & & \rho(x; K_n) & \xrightarrow{\text{pointwise}} & \rho(x; K) \end{array}$$

6.1 A non-recursive method

The above theorem has a fundamental curse: it involves functions defined recursively, and recursion is not only better to be avoided, but it is also incompatible with the programming language we wish to implement our results in. Therefore we need to circumvent recursion and provide a “closed form” for our solution.

Definition 6.5 Let $\{f_1, \dots, f_n\}$ be a family of functions. We define the composition operator to be:

$$\bigcirc_{i=1}^n f_i(x) := (f_n \circ f_{n-1} \circ \dots \circ f_1)(x).$$

Theorem 6.6 Let $d_n(x)$ be the sequence defined recursively via:

$$d_n(x) \sim \min_{1 \leq i \leq m} \{d_{n-1}(f_i^{-1}(x))\} \quad d_0(x) = \rho(x; K_0)$$

Where $\{f_1, \dots, f_m\}$ and K_0 are defined as in the above theorem. Fix x , then:

$$d_n(x) \sim d_0 \left(\bigcirc_{i=1}^n f_{\mu_i}^{-1}(x) \right)$$

Where each μ_j , which we refer to as the j^{th} minimising index, is such that:

$$\min_{1 \leq i \leq m} \{d_{j-1}(f_i^{-1}(x))\} = d_{j-1}(f_{\mu_j}^{-1}(x))$$

Proof. The first term of the sequence corresponds to

$$d_1(x) = \min_{1 \leq i \leq m} d_0(f_i^{-1}(x))$$

The quantity $d_0(f_i^{-1}(x))$ gets minimised for some μ_1 , i.e:

$$\min_{1 \leq i \leq m} d_0(f_i^{-1}(x)) = d_0(f_{\mu_1}^{-1}(x))$$

We now prove the claim by induction. The base case holds by definition of μ_1 . Now suppose that the claim holds for $n \leq k-1$. We now see that

$$d_k(x) \sim \min_{1 \leq i \leq m} \{d_{k-1}(f_i^{-1}(x))\} \sim \min_{1 \leq i \leq m} \left\{ d_0 \left(f_i^{-1} \circ \bigcirc_{j=1}^{k-1} f_{\mu_j}^{-1}(x) \right) \right\}$$

This last equivalence is due to Corollary 5.27. Now finally, we observe that by definition this last quantity is equal to

$$d_0 \left(f_{\mu_k}^{-1}(x) \circ \bigcirc_{j=1}^{k-1} f_{\mu_j}^{-1}(x) \right) = d_0 \left(\bigcirc_{j=1}^k f_{\mu_j}^{-1}(x) \right)$$

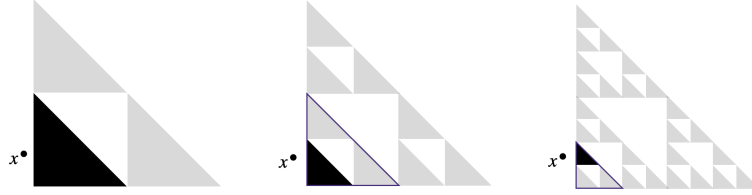
As required. ♡

We can thus express our final result as:

$$\rho(\bigcirc_{j=1}^n f_{\mu_j}^{-1}(x); K_0) \sim d_n(x) \xrightarrow{\text{pointwise}} d(x) \sim \rho(x; K)$$

Remark 6.7 The minimising indices μ_j depend on x , therefore the minimising calculations must be performed every time $d_0(\bigcirc_{i=1}^n f_{\mu_i}^{-1}(x))$ wants to be computed.

Example 6.8 (Illustration) Here we see a heuristic example of this theorem to obtain an equivalent distance function to the n^{th} forward iterate of the IFS whose attractor is the Sierpinski Triangle.



Three forward iterates are shown: $T[K_0]$, $T^{\circ(2)}[K_0]$ and $T^{\circ(3)}[K_0]$. Where T is the Hutchinson operator of the IFS and each f_i corresponds to a homothety towards one of the three vertices of the triangle. In each iterate, the region whose “index” minimises the distance to x is shaded. The theorem states that to find the region of the next iterate whose “index” minimises the distance to x , it is enough to look among the subdivisions of the shaded region of the previous iterate.

We now present pseudo-code for computing the n^{th} term of the sequence of equivalent distance functions:

Algorithm 3 Compute $d_n(x)$

Input: $x \in S$, $n \in \mathbb{N}$

for $i = 1, \dots, n$ **do**

 Calculate the index μ_i such that $\rho(f_{\mu_i}^{-1}; K)$ is the smallest.

 Set $x \leftarrow f_{\mu_i}^{-1}(x)$.

end for

return $\rho(x; K_0)$.

As we can see, this method resembles the idea of the Region Subdivision algorithm presented in section 4.1.2, because in some sense, both methods perform a search over subdivisions of the fractal. The main difference is that the region subdivision algorithm uses the maps $\{f_1, \dots, f_m\}$ of the IFS, whereas the equivalent distance function method uses the inverses.

7 Final comparisons

We have now seen how, given an IFS $\{f_1, \dots, f_m\}$ with attractor $K = \lim_{n \rightarrow \infty} T^{\circ(n)}[K_0]$, one can Ray-March an approximation to K . The ideas we have discussed are divided into two categories: firstly estimating the distance function itself to later Ray-March the attractor (Section 4), or directly finding an equivalent distance function that produces the same Ray-Marching as the true distance function (Section 6). We argued that out of the methods in Section 4, only the Region-Subdivision algorithm is a good candidate for the implementation of the Ray-Marching algorithm. To conclude the discussion of this project, we will see how using equivalent distance functions compares to the Region-Subdivision method. We will refer to these methods as EDF and RS methods respectively.

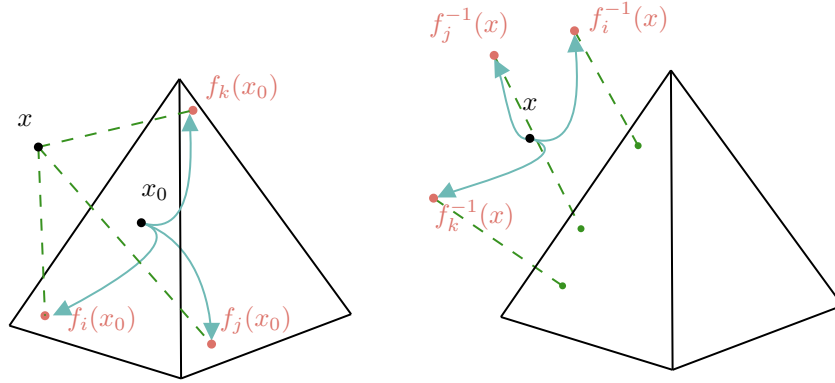


Figure 30: Conceptual difference between the Region-Subdivision method (left) and Equivalent-Distance functions (right) when estimating the distance to a point x : the RS method starts with a point x_0 inside the fractal and applies the IFS to this point. We then aim to minimise the distance from x to these new points. The EDF Method applies the inverses of the IFS maps to x . We then aim to minimise the distance from the set to each of these new points.

7.1 Practicality

Both methods are algorithmically very similar, so none of the two methods is particularly easier to implement in code than the other. However there is a major difference on which kinds of IFS each method can compute. As explained in Remark 6.4, to compute an equivalent distance function to the attractor, we required our transformations to be invertible regular contractions as opposed to just contractions. This is a major constraint that the RS algorithm does not require.

7.2 Time complexity

Let us now compare how efficient the algorithms are. Recall that Region-Subdivision returns an arbitrarily accurate approximation to the distance function itself, whereas equivalent-distance functions don't. However, we can note that the final intent of both algorithms is to Ray-March a set K_ϵ , that is less than ϵ Hausdorff-distance away from K . We can compare the time complexity of both algorithms as we take $\epsilon \rightarrow 0$. Recall that when we say *Ray-March a set*, we say it in view of Definition 5.6

The first thing to note is that each algorithm has to be performed a certain amount of times before reaching the desired tolerance. The performed operations in each iteration are the following:

N^{th} iteration of RS Method	N^{th} iteration EDF Method
for $i = 1, \dots, m$ do	for $i = 1, \dots, m$ do
Calculate $x_i = f_i(x_{n-1})$	Calculate $x_i = f_i^{-1}(x_{n-1})$
end for	end for
Calculate the index j such that $\rho(x, x_j)$ is minimised.	Calculate the index j such that $\rho(x_j; K_0)$ is minimised.
Set $x_n = x_j$	Set $x_n = x_j$

Once again we see the similarity of both algorithms, and assuming that the time costs of calculating $f_i(x)$, $f_i^{-1}(x)$, $\rho(x, y)$ and $\rho(x; K_0)$ are all equal, we can conclude that the only aspect that decides the time cost of each algorithm is the growth of iterations needed as we let $\epsilon \rightarrow 0$. We will now calculate the number of iterations each algorithm requires to get within ϵ Hausdorff distance of the attractor and compare the behaviour of both expressions. For simplicity, suppose all the contractivity factors r_i are equal to r .

7.2.1 Region-Subdivision

To find the number of required iterations, we simply recall the proof of Proposition 4.2 and see that this algorithm terminates once we have covered K by balls of ϵ -radius: starting with some ball A centered at some $x_0 \in K$ with a large enough radius to contain the entire fractal and applying the transformations to each of the successive balls. Note that $\text{diam}(f_i(A)) = r \text{diam}(A)$, so if we require this last quantity to be less than or equal to some $\epsilon > 0$, we deduce that the amount of iterations, n , required to complete the algorithm satisfies:

$$n \geq \frac{\log(\epsilon)}{\log(r)} - \frac{\log(\text{diam}(A))}{\log(r)}$$

Remark 7.1 This inequality is sharp.

7.2.2 Equivalent Distance Function

It will be slightly more involved to find bounds on the number of iterations for this method. Let us first present a lemma:

Lemma 7.2 Let (x_n) be a Cauchy sequence in a complete metric space (S, ρ) with limit $x \in S$. If a natural number $N > 0$ is such that for any $m \geq n \geq N$ one has $\rho(x_n, x_m) < \epsilon$ for some $\epsilon > 0$, then $\rho(x_k, x) < \epsilon$ for any $k \geq N$.

Proof. Suppose $x \notin \mathbb{B}_\epsilon(x_N)$. Let $0 < \lambda = \rho(x, x_N) - \epsilon$. By definition of convergence, we have that for values of n large enough, the elements x_n of the sequence are in $\mathbb{B}_\lambda(x)$. Contradicting the construction of N in the definition of Cauchy sequence. $\heartsuit$

Theorem 7.3 Let $\{f_1, \dots, f_m\}$ be a regular IFS with attractor $K = \lim_{n \rightarrow \infty} K_n \equiv \lim_{n \rightarrow \infty} T^{\circ(n)}[K_0]$, where T is the Hutchinson operator. Let r be the contractivity factor of the map T and let (d_n) be a sequence of functions such that $d_n(x) \sim \rho(x; K_n)$ as described in Theorems 6.2 and 6.6. Then given any $\epsilon > 0$, if:

$$n \geq \frac{\log(\epsilon)}{\log(r)} + \frac{\log(1-r)}{\log(r)} - \frac{\log(d_{\mathbb{H}}(K_0, K_1))}{\log(r)}$$

Then one has $d_{\mathbb{H}}\left(\bigcap_{\delta>0} \mathcal{R}_{d_n}(\delta), K\right) < \epsilon$, where $\mathcal{R}$ is the Ray–Marching function described in section 5.1

Proof. First make the observation that for any set $\Sigma \in \mathbb{H}(S)$, one has $\Sigma = \bigcap_{\delta>0} \mathcal{R}_{\rho(x;\Sigma)}(\delta)$. Now we note that:

$$d_{\mathbb{H}}\left(\bigcap_{\delta>0} \mathcal{R}_{d_n}(\delta), K\right) = d_{\mathbb{H}}\left(\bigcap_{\delta>0} \mathcal{R}_{\rho(x;K_n)}(\delta), K\right) = d_{\mathbb{H}}(K_n, K)$$

Where the first equality is a consequence of d_n being equivalent to $\rho(x; K_n)$. Thus if we want to make $d_{\mathbb{H}}(\bigcap_{\delta>0} \mathcal{R}_{d_n}(\delta), K) < \epsilon$, it suffices to make $d_{\mathbb{H}}(K_n, K) < \epsilon$. By the previous lemma, it suffices to find an N such that for any $m \geq n \geq N$ we have $d_{\mathbb{H}}(K_n, K_m) < \epsilon$. By following the proof of Banach's Fixed Point Theorem applied to the sequence (K_n) in the space $(\mathbb{H}(S), d_{\mathbb{H}})$, we see that for any $m \geq n$:

$$d_{\mathbb{H}}(K_n, K_m) \leq \frac{r^n}{1-r} d_{\mathbb{H}}(K_0, K_1)$$

Now imposing this second quantity to be less than ϵ and rearranging yields the desired result. $\heartsuit$

Let us now compare the bounds on the Region-Subdivision against the Equivalent Distance Function method

Region-Subdivision

$$n \geq \frac{\log(\epsilon)}{\log(r)} - \frac{\log(\text{diam}(A))}{\log(r)}$$

Equivalent Distance Functions

$$n \geq \frac{\log(\epsilon)}{\log(r)} + \frac{\log(1-r)}{\log(r)} - \frac{\log(d_{\mathbb{H}}(K_0, K_1))}{\log(r)}$$

The first observation to make is that in both cases, the amount of iterations one needs to do to Ray-March a set K_ϵ that is within ϵ -tolerance of K grows like $\mathcal{O}(\log(\epsilon))$. For other applications, this type of time complexity analysis would suffice, but there is an important observation to make here, the constants that appear in each of the bounds **do** matter. The first reason is that due to the nature of Ray-Marching, for each extra iteration that we demand, the total running time increases substantially (To Ray-March a 1080p image, the computer needs to estimate distances several million times. Slight increases in computation time per calculation add up quickly). The second reason is that in practice we will not take $\epsilon \rightarrow 0$. Indeed: for all intents and purposes, one cannot tell the difference between the images of two sets whenever their Hausdorff distance is less than $\epsilon \leq 0.001$. As a result, every slight improvement on the required iterations is welcome, and as one can observe in the formulas above, the EDF method allows us the freedom to choose a set K_0 that makes $d_{\mathbb{H}}(K_0, K_1)$ small. This might seem insignificant, but let us look at a concrete example:

The recurrent example in this project was Menger's Sponge. For this specific fractal, the contractivity ratio is of $r = 1/3$ and the diameter of the fractal is $\sqrt{3}$. Thus the optimal choice of A for the RS Method is a ball of said diameter. On the other hand, the EDF method allows the freedom of choice of K_0 . A natural choice of K_0 is a cube of side-length 1, and it is easily shown that $d_{\mathbb{H}}(K_0, K_1) = 1/6$. We can now use these values and view in Figure 31 how the number of iterations (vertical axis) changes as ϵ (horizontal axis) gets closer to 0. Some values are presented in Table 1

ϵ	n (EDF)	n (RS)
0.1	1	3
0.01	3	4
0.001	5	6
0.0001	6	8

Table 1: Error tolerance against iterations needed

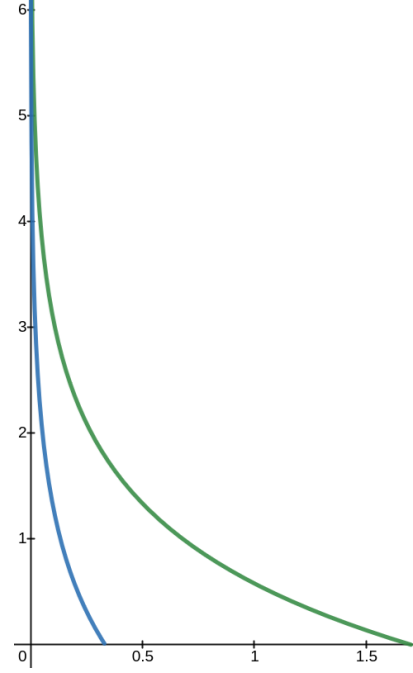


Figure 31: Amount of required iterations to get within ϵ tolerance. In green (above) is the curve for the RS method and in blue (below) is the curve for the EDF method.

7.2.3 Quality of images

The fact that the EDF method achieves a good approximation faster than the RS method should already suggest that the quality of the images are better, but the main point in this comparison is that sometimes it is equally valuable to visualise the first iterations of an IFS as it is to visualise a set that resembles its attractor. The EDF method does this indisputably better than the RS method. The reason for this is that each IFS has a “preferred” starting set K_0 : for the Menger’s Sponge it is a cube, for Sierpinski’s Triangle, a tetrahedron or any other pyramid-like shape. This means that in the EDF method, the Ray-Marching of each iteration gives a better illustration of the process of converging to the attractor. To illustrate this point, compare the pictures in Figure 13 and in Figure 33. In both illustrations we see the results of Ray-Marching the third, fifth and eighth iterations.

Another interesting point is that since the EDF method allows for freedom to choose K_0 , one can play around and visualise how the IFS behaves with different starting sets. We know by the Attractor Theorem that this choice is irrelevant, since everything eventually converges to the attractor, but this is another feature of the EDF method that cannot be done by the RS method. See Figure 32 for a fun demonstration of this concept. Moreover, the EDF method allows for the visualisation of the intersection of sets and even complements (provided that you are willing to sacrifice an EDF in return of an underestimator).

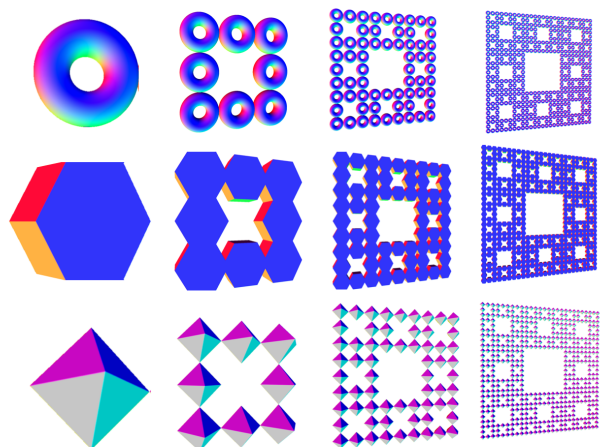


Figure 32: Ray-Marching of K_0 through K_3 , for three different choices of K_0 .

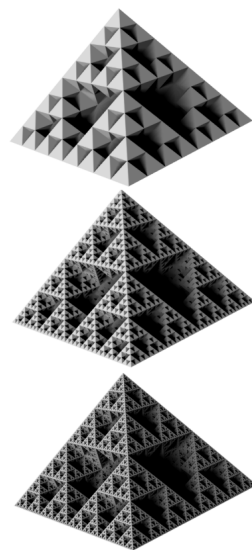


Figure 33: Iterations 3, 5 and 8 of the Sierpinski “pyramid”.

8 Acknowledgements

I would like to give special thanks to the following people: Dr. Jean Lagacé, who agreed to work with me on this project even when I did not have a clear picture of what it would turn out to look like, and who helped me turn a few scattered ideas into what I hope is some robust work. I have learnt many valuable lessons from him that have helped me mature as a Mathematician. I would also like to thank my dear friend, flatmate, and fractal enthusiast Sam Tomlin, who among other things: maintained with me **endless** hours of conversations about this project, constantly pushed my creativity, and enthusiastically listened to me while I complained once again to him about convexity or some inequality I desperately needed but could not obtain, sometimes even before having had breakfast.

9 Appendix

9.1 Code to Ray-March a Fractal with the EDF method

```
1
2 //This is a working example developed in the IDE KodeLife
3 #version 150
4
5 uniform vec2 resolution;
6
7 //We define the fixed parameters needed for the Ray-Marching Algorithm
8 #define MAX_ITER 400
9 #define MAX_DIST 200
10 #define TOLERANCE 0.00001
11
12
13 //Ignore the following few lines
14 in VertexData
15 {
16     vec4 v_position;
17     vec3 v_normal;
18     vec2 v_texcoord;
19 } inData;
20
21 out vec4 fragColor;
22
23 void rot(inout vec2 pos, in float a){
24     //This is a function used to rotate the camera later
25     float c = cos(a);
26     float s = sin(a);
27     pos = vec2(pos.x*c+pos.y*s,-pos.x*s+pos.y*c);
28 }
29
30
31 float sdOctahedron(vec3 p, float s)
32 // Primitive Distance Function to an Octahedron. Code obtained from
33 // Inigo Quilez's website [Quia]
34 {
35     s = 1;
36     p = abs(p);
37     float m = p.x + p.y + p.z - s;
38     vec3 r = 3.0*p - m;
39
40     #if 0
41
42     vec3 o = min(r, 0.0);
43     o = max(r*2.0 - o*3.0 + (o.x+o.y+o.z), 0.0);
44     return length(p - s*o/(o.x+o.y+o.z));
45     #else
46
47     vec3 q;
48     if( r.x < 0.0 ) q = p.xyz;
49     else if( r.y < 0.0 ) q = p.yzx;
50     else if( r.z < 0.0 ) q = p.zxy;
```

```

51     else return m*0.57735027;
52     float k = clamp(0.5*(q.z-q.y+s),0.0,s);
53     return length(vec3(q.x,q.y-s+k,q.z-k));
54 #endif
55 }
56
57 //Distance function of K_0
58 float distanceEstimator(in vec3 p){
59     float s1 = sdOctahedron(p,0.9);
60     return s1;
61 }
62 }
63
64 //Inverse map of each f_i
65 vec3 inverse_contract(in vec3 pos, in float ratio, in vec3 center){
66     return (pos+ratio*center)/ratio;
67 }
68
69 }
70
71
72 //We declare the points to which each map in the IFS contracts space towards
73 vec3 centrs[6] = vec3[6](vec3(0,-1,0),vec3(-1,0,0),vec3(0,0.,1),vec3(0,1,0),vec3(1,0,0),vec3
    (0,0,-1));
74
75
76 //Number of iterations to perform the computation of the Equivalent Distance Function
77 const int max_iters =4;
78
79
80 //Equivalent Distance Function
81 float EDF(in vec3 p){
82
83     int min_indexes[max_iters];
84     float ratio = 0.5;
85     vec3 composed_pos;
86     vec3 new_pos;
87
88
89
90     for (int iteration = 0; iteration <= max_iters - 1; iteration++){ //main loop that runs
        N iterations
91         for (int k = 0; k <= iteration; k++){ // loop that finds the composition of the k-1
            st inverse IFS of pos
92             if (k==0){ // if k=0 we can't access the array as it is empty still
93                 composed_pos = p;
94
95             } else {
96
97                 composed_pos = inverse_contract(composed_pos,ratio,centrs[min_indexes[k-1]])
98             }
99         }
100     }

```

```

101     // now we have found the new_position that we need to iterate in order to find the
minimizing index
102     float min_dist = 1000;
103     int min_index = 1;
104     for (int j = 0; j <= 5; j++){
105         vec3 new_pos = inverse_contract(composed_pos, ratio, centrs[j]);
106         if (distanceEstimator(new_pos) < min_dist){
107             min_dist = distanceEstimator(new_pos);
108             min_index = j;
109         }
110     }
111     // now we have found the minimizing index for the current iteration, we label it as
such
112     // and populate the array
113     min_indexes[iteration] = min_index;
114
115 }
116 // compose all inverses and calculate final distance  $x' = \bigcirc_{i=1}^{\max\_iters} f_i^{-1}(x)$ 
117 vec3 final_position = p;
118 for (int i = 0; i <= max_iters - 1; i++){
119     final_position = inverse_contract(final_position, ratio, centrs[min_indexes[i]]);
120 }
121
122 return distanceEstimator(final_position)*pow(ratio,max_iters);
123 }
124
125 //Distance value which we feed into the Ray-Marching Algorithm
126 //Here is the point where you can take unions, intersections, etc.
127 float final_dist(in vec3 pos){
128     return EDF(pos);
129     //return sdOctahedron(pos,0.);
130 }
131 }
132
133 //Ray-Marching Algorithm
134
135 float rayMarch(in vec3 ro, in vec3 rd){
136
137     float t = 0.0;
138     float d;
139     float ratio = 1./2.;
140
141     for (int i = 0; i < MAX_ITER; i++){
142         vec3 pos = ro + t*rd;
143         d = final_dist(pos);
144         //Here we use the modified conditions of the Ray-Marching Algorithm required for the
EDF method to work
145         //Instead of checking if d < TOLERANCE || d > MAX_DIST, we scale the tolerances by the
similarity constants of the EDF
146         if (d < TOLERANCE*pow(ratio,max_iters) || d > MAX_DIST/pow(ratio,max_iters)){
147             break;
148         }
149         else{

```

```

150         t+= d ;
151     }
152 }
153 return t;
154 }
155
156 // Extra code to achieve the shading and lighting. Obtained from Inigo Quilez [Quid]
157
158 float diffuse(in vec3 nor, in vec3 ld){
159     return max(dot(nor,ld),0.0);
160 }
161
162 float softshadow( in vec3 ro, in vec3 rd, float mint, float maxt, float k )
163 {
164     float res = 3.;
165     float t = mint;
166     for( int i=0; i<256*2 && t<maxt; i++ )
167     {
168         float h = final_dist(ro + rd*t);
169         if( h<0.001 )
170             return 0.0;
171         res = min( res, k*h/t );
172         t += h;
173     }
174     return res;
175 }
176
177
178
179 //Rendering Algorithm, this is the Algorithm in charge of coloring each pixel on screen
180
181 vec3 render(in vec3 ro, in vec3 rd){
182
183
184     float t = rayMarch(ro,rd);
185     vec3 pos = ro + t*rd;
186
187     //extra code to make the shadows work, again obtained from Inigo Quilez
188     vec3 lig = normalize( vec3(0.2, 0.8, 0.3) );
189     float dif = softshadow(pos,lig,0.01, 4.0, 0.1);
190     float s = dif*10;
191
192     //Here we color a pixel based on "background color" or "set color"
193     if (t < MAX_DIST){
194
195         return vec3(s,0.2,0.4);
196     }else{
197         return vec3(1,1,1);
198     }
199 }
200
201
202 void main(void)
203 {

```

```

204 // we obtain each pixel on screen, this is given by the
205 //inData.v_texcoord object. We store it in a vector p
206 vec2 p = -1. + 2. * inData.v_texcoord;
207
208 //The vector is rescaled for aesthetic purposes
209 p.x *= resolution.x/resolution.y;
210 p*=1.7;
211 vec3 ro = 1.5*vec3(1.,3.1,0.);
212 rot(ro.xz,.6);
213 rot(ro.zy,0.4);
214 rot(ro.xy,-0.2); //0.6
215
216
217
218 //The Ray-Direction is calculated, some miscellaneous code
219 // is added for aesthetic purposes. The following 5 lines are a typical example of a
    camera setup
220 // the code was obtained from Paul Houx's shaderToy Cinder example
221
222 float f = 1.2;
223 vec3 ww = f*normalize(vec3(0.0,0.0,0)-ro);
224 vec3 uu = normalize(cross(vec3(1.1,2,0),ww));
225 vec3 vv = normalize(cross(ww,uu));
226 vec3 rd = normalize(p.x*uu+p.y*vv+4.8*ww);
227
228
229 // The color of pixel is rendered
230 vec3 col = render(0.92*ro-vec3(0.,0.2,0.53),rd);
231
232 // The pixel is colored
233 fragColor = vec4(col,1);
234 }

```

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